Math 114 Class notes
Fall 2026

To accompany
Precalculus 2e
by *Abramson*

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Math 114 Formula Sheet

Unit One

Fractional exponents: $x^{a/b} = \sqrt[b]{x^a}$

Subtracting exponents: $x^{a-b} = \dfrac{x^a}{x^b}$

Average Rate of Change: $\dfrac{f(b) - f(a)}{b - a}$

Slope of a Line: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

Point-Slope form: $y - y_1 = m(x - x_1)$

Slope-Intercept form: $y = mx + b$

Unit Two

General form: $ax^2 + bx + c = 0$

Standard form: $a(x - h)^2 + k = 0$

Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

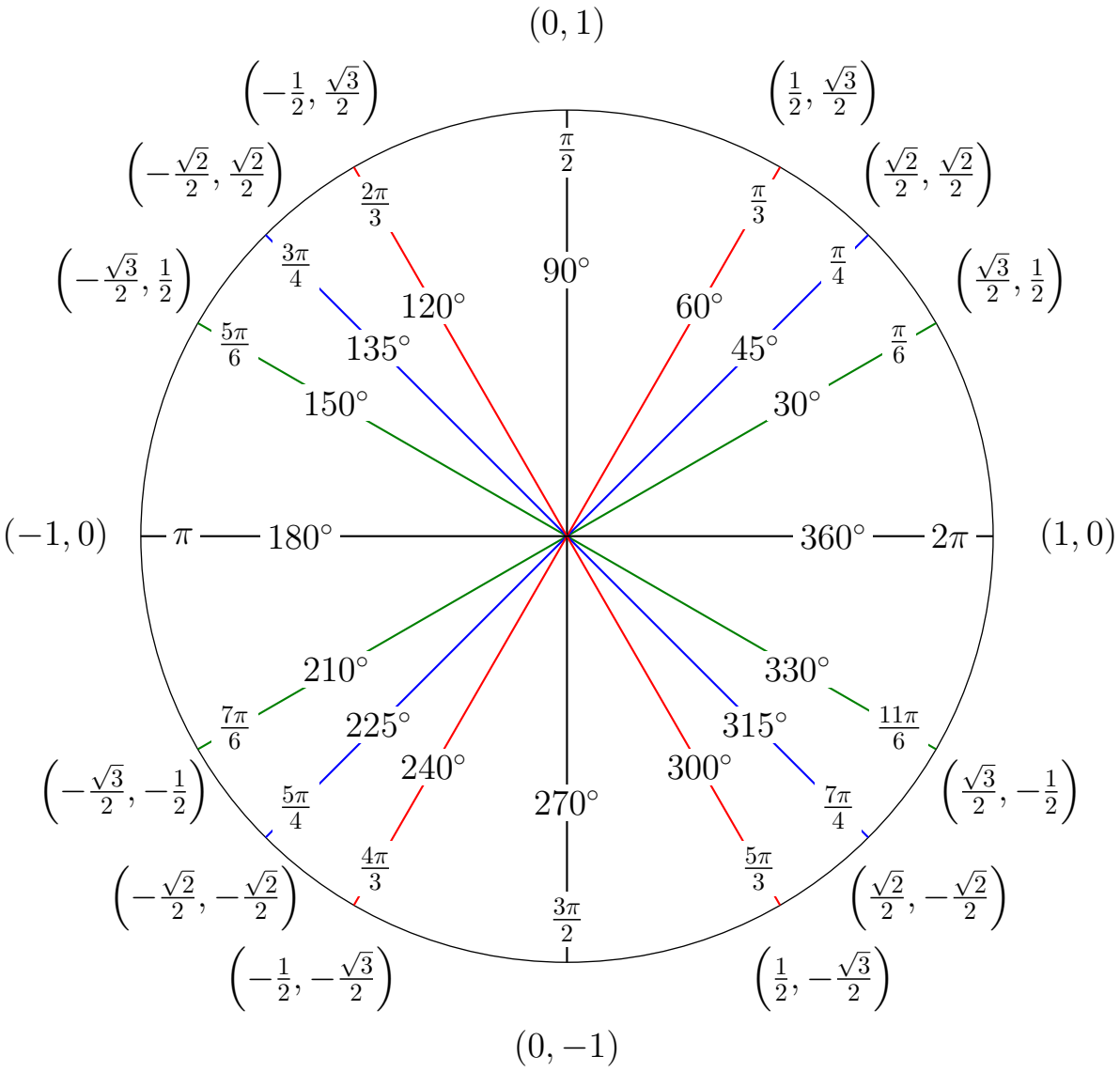
Vertex of a parabola: (h, k) where $h = -\dfrac{b}{2a}$, $k = f(h) = c - \dfrac{b^2}{4a}$

Unit Three

Pythagorean Identity: $\cos^2(x) + \sin^2(x) = 1$

Unit Circle

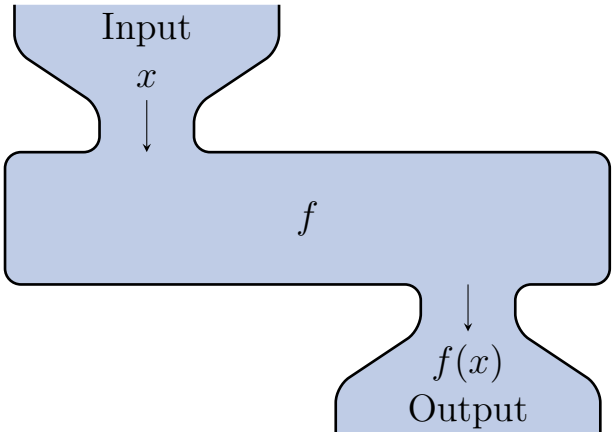
$$\big(\cos(\theta),\ \sin(\theta)\big)$$

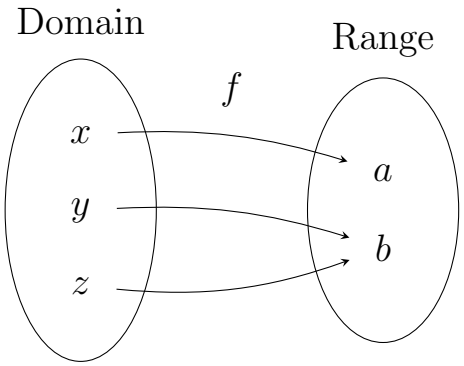


$$180^\circ = \pi \text{ rad}$$

1.1: Functions and Function Notation

Definition.
A **function** is a relation in which each possible input value leads to exactly one output value.
The input values make up the **domain**, and the output values make up the **range**.





Example. Let $f(x) = 2x^2 - 2x + 1$. Evaluate the following

$f(1)$ $f(-2)$ $f(a)$

$f(a+h)$ $\dfrac{f(a+h)-f(a)}{h}$

Example. Suppose that the following tables represent a carnival truck's menu prices. Which table would represent a valid function?

Item	Price
Funnel cake	\$8
Chicken strips	\$9
Ribbon fries	\$7
Cheese nachos	\$6

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$7
Cheese nachos	\$6

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$8
Funnel cake	\$6

Example. Based on an old match-makers rule, the socially "acceptable" minimum age of someone you can date is half your age plus seven. Make a function that takes your age, x, and finds the minimum age of someone you can date, y.

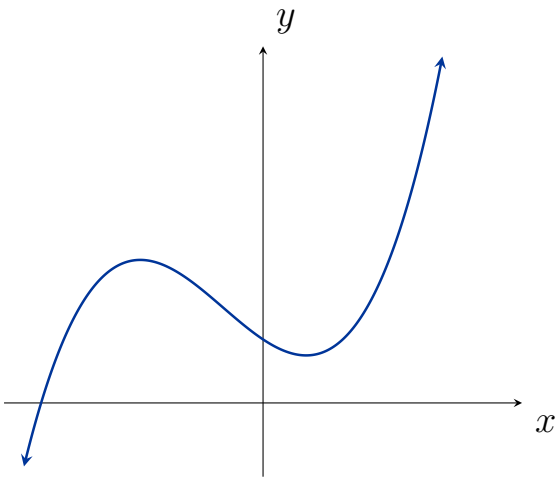
Example. Create a table to represent a function that takes each month and returns the number of days (in a non-leap year).

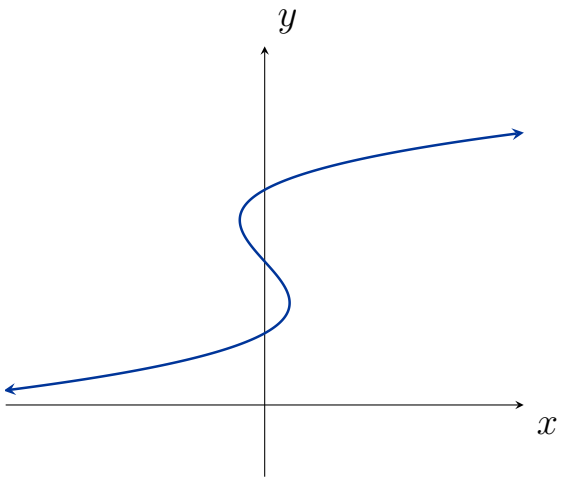
Example. If possible, express the relationship $2n + 6p = 12$ as a function $p = f(n)$. Graph the result.

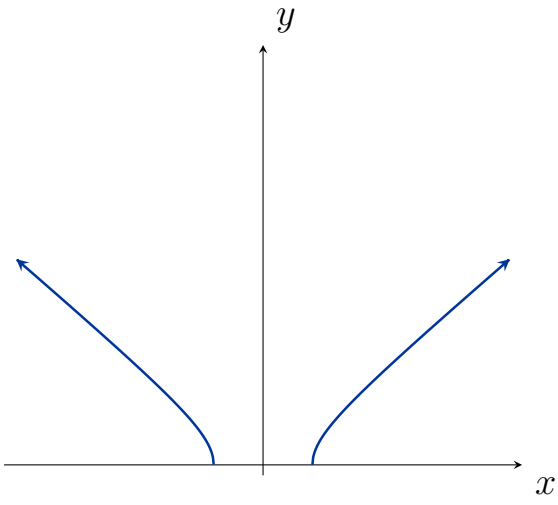
Example. If possible, express the relationship $x^2 + y^2 = 1$ as a function $y = f(x)$. Graph the result.

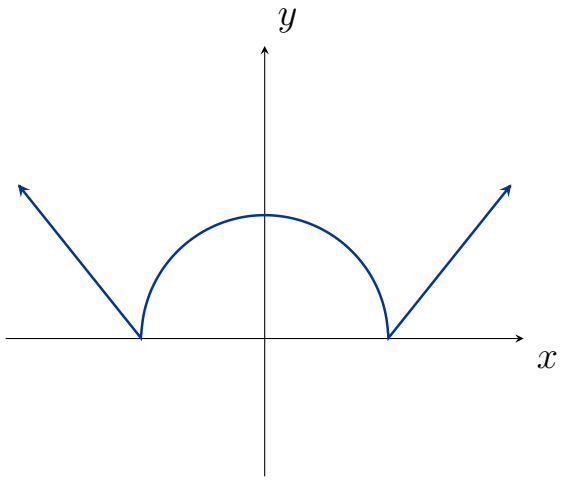
Definition. (Vertical Line Test)
A curve in the xy-plane is the graph of a function $y = f(x)$ (an explicit function) if and only if each vertical line intersects it in at most one point

Example. Use the vertical line test on the following graphs to determine which graphs may represent an explicit function:









Definition. (One-to-One)
The **horizontal line test** determines if every horizontal line passes through the graph of a function in at most one point.
A function is said to be **one-to-one** if each output value corresponds to *exactly* one input value (and hence passes the horizontal line test).

Example. Which of the following represent a one-to-one function? Graph

$y = x^2$

$y^2 = x$

$f(x) = \sqrt{x}$

$y = |x|$

$f(x) = (x+1)(x-1)(x-2)$

The equation of a circle:
$x^2 + y^2 = 1$

The area of a circle:
$A = \pi r^2$

Graphs of basic functions

1.2: Domain and Range

Definition. (Interval notation)
We can specify the domain and/or range of a function using interval notation:

- Strict inequalities (exclusive) are represented using parenthesis:

$$x > 3 \iff (3, \infty) \qquad\qquad x < 9 \iff (-\infty, 9)$$

  Note: Parenthesis are **always** used when an interval's end points are infinite

- Nonstrict inequalities (inclusive) are represented using square brackets:

$$7 \leq x \leq 11 \iff [7, 11]$$

- Disjoint intervals can be unioned together:

$$x \neq -2 \iff (-\infty, -2) \cup (-2, \infty)$$

- Any combination of the above can be used:

$$[-1, 1] \cup (2, 3] \cup (4, 5) \cup [6, \infty)$$



Recall that the domain is the set of all *possible* input values.

Example. Find the domain of the following functions (use interval notation): Graphs

$f(x) = x - 3$

$A = \pi r^2$

$y = \sqrt{x-1}$

$y = \dfrac{x+2}{x^2-4}$

$y = \sqrt[3]{x-4}$

$f(x) = \dfrac{\sqrt[4]{x^2+4}}{(x+1)\sqrt{x-1}}$

Example. Identify the domain and range of the following: Graphs

$x(x+1)(x+2)^2, \quad x \geq -2$

$\sin(x)$

$\log(x)$

$2^x, \quad x \geq -1$

Definition.
A **piecewise** function is a function with different definitions for different portions of the domain.

Example. Rewrite the following as piecewise functions:

$|x| =$

$\dfrac{x}{|x|} =$

$|x-1| + |4-x| =$

Example. Graph the following piecewise function: Graphs

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$

1.3: Rates of Change and Behavior of Graphs

Definition. (Rate of Change)
A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are "output units per input units". The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

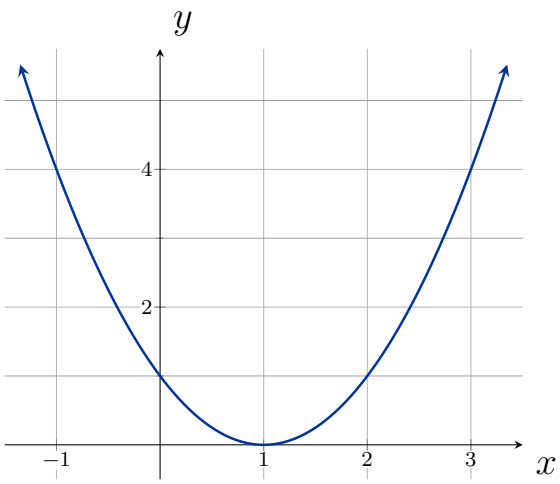
Example. Suppose the following table contains temperatures recorded through the day.

h	7 AM	9 AM	11 AM	1 PM	3 PM
$T(h)$	$78°F$	$81°F$	$87°F$	$93°F$	$98°F$

Find the average rate of change between

7 AM and 9 AM 9 AM and 11 AM 7 AM and 3 PM

Example. Using the following graph, determine the average rate of change on the interval $[-1, 2]$



Example. Find the average rate of change of $f(x) = x^2 - \dfrac{1}{x}$ on the interval $[2, 4]$.

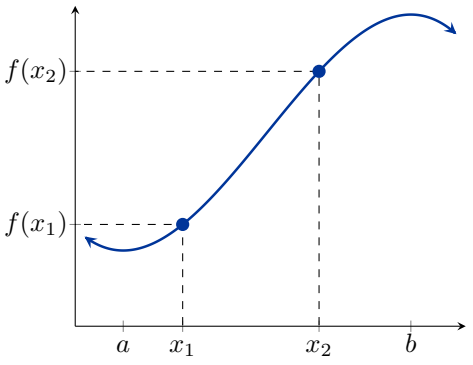
Example. Find the expression for the average rate of change of $g(t) = t^2 + 3t + 1$ on the interval $[0, a]$.

Example. Find the expression for the average rate of change of $j(t) = \dfrac{1}{t+3}$ on the interval $[1, 1+h]$.

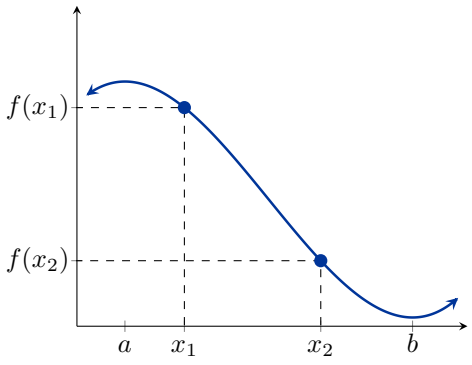
Example. Find the expression for the average rate of change of $g(x) = \sqrt{x-1}$ on the interval $[x, x+h]$.

Definition.
Consider the function $f(x)$ on the interval (a, b). Given *any* two numbers x_1 and x_2 in (a, b) where $x_1 < x_2$, we say f is

increasing if $f(x_1) < f(x_2)$

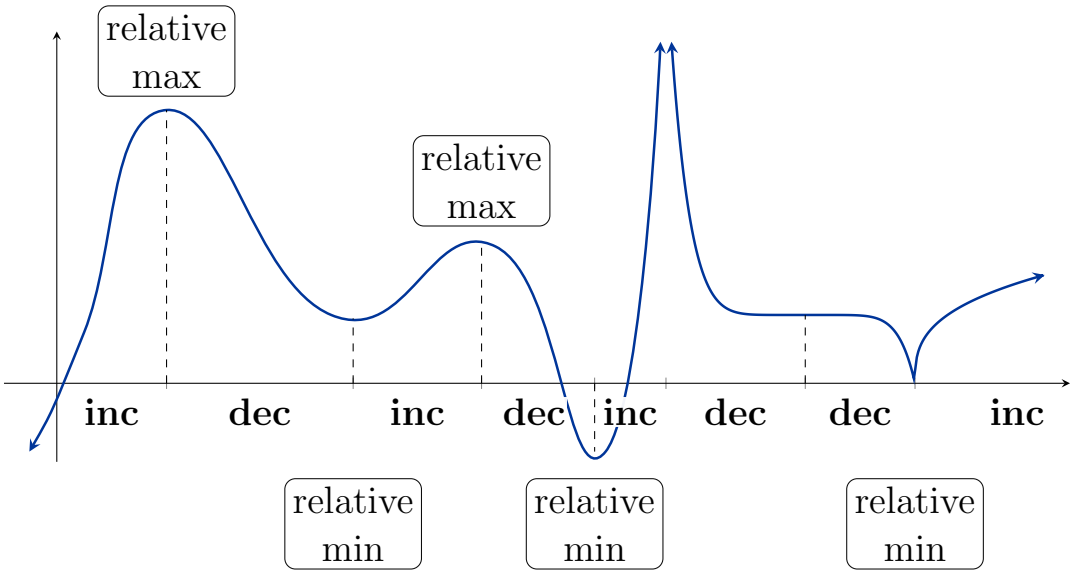


decreasing if $f(x_1) > f(x_2)$



Definition. (Relative/Local Extrema)
A function f has a

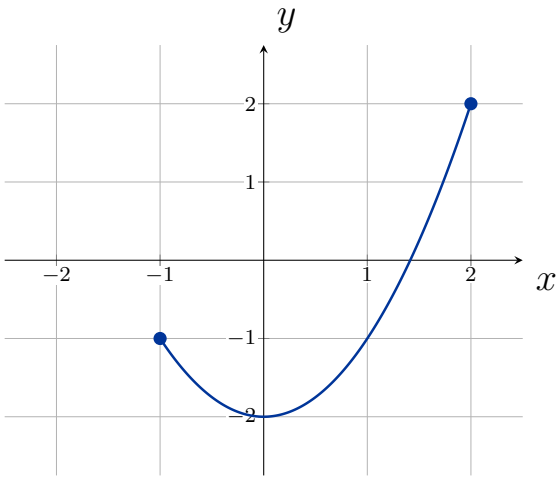
- **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in (a, b)
- **relative minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in (a, b)



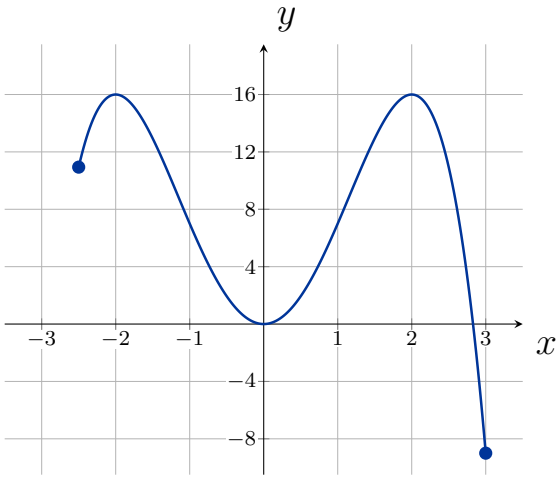
Definition. (Absolute Extrema)
A function f defined on a domain D has an

- **absolute maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in D
- **absolute minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in D

Example. Using the following graph, find any relative and absolute extrema.



Example. Using the following graph, find any relative and absolute extrema.



1.4: Composition of Functions

Definition. (The Composition of Two Functions)
Let f and g be functions. Then the composition of g and f is the function $g \circ f$ defined by

$$(g \circ f)(x) = g(f(x))$$

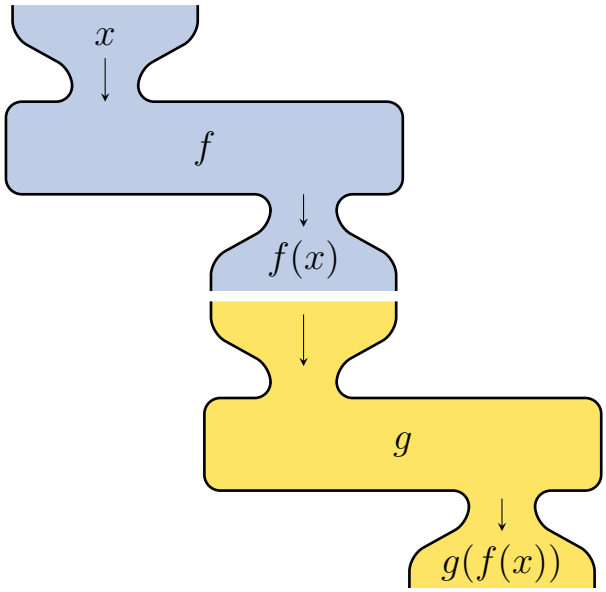
The domain of $g \circ f$ is the set of all x is the domain of f such that $f(x)$ lies in the domain of g.

Example. Let $f(x) = \sqrt{x+1}$ and $g(x) = 4 - x$. Find the domain of the following:

$g(f(x)) =$

$f(g(x)) =$

$f(f(x)) =$



Example. For the following, find functions $g(x)$ and $h(x)$ so that $f(x) = g(h(x))$:

$f(x) = \dfrac{1}{x+2}$

$f(x) = \sqrt{x-4}$

$f(x) = \left(x^2 + 3x - 8\right)^4$

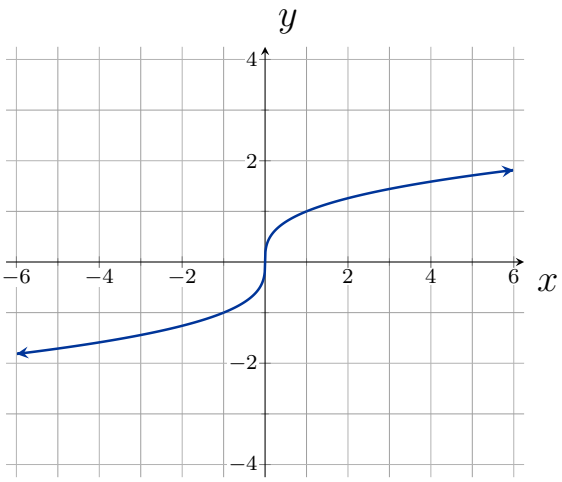
$f(x) = \dfrac{x-3}{\sqrt{-(x-3)}}$

1.5: Transformation of Functions

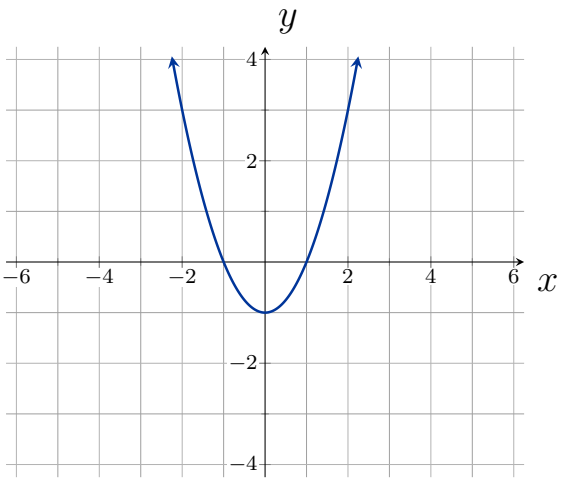
Definition. (Vertical Shift)
Given a function $f(x)$, a new function $g(x) = f(x) + k$, where k is a constant, is a **vertical shift** of the function $f(x)$. All the output values shift vertically by k units (**up** if $h > 0$, **down** if $h < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

Graph $f(x) + 2$:



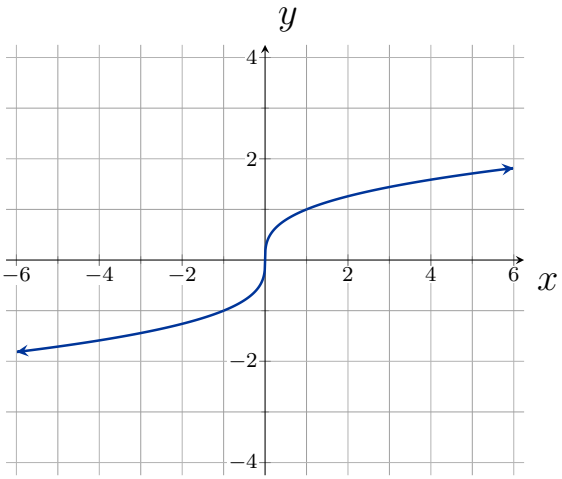
Graph $f(x) - 1$:



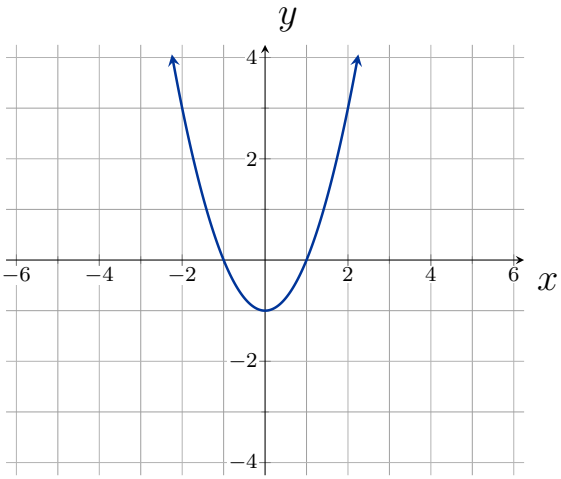
Definition. (Horizontal Shift)
Given a function $f(x)$, a new function $g(x) = f(x - h)$, where h is a constant, is a **horizontal shift** of the function $f(x)$. All the output values shift horizontally by h units (**right** if $k > 0$, **left** if $k < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

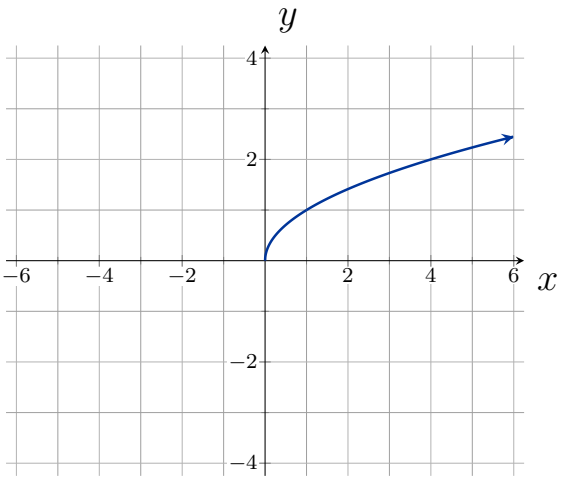
Graph $f(x + 2)$:



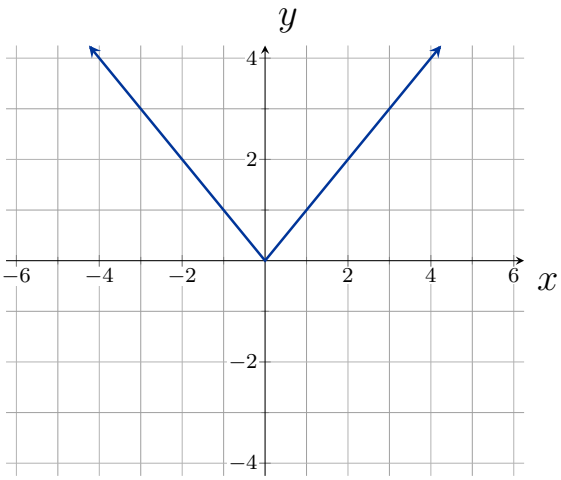
Graph $f(x - 1)$:



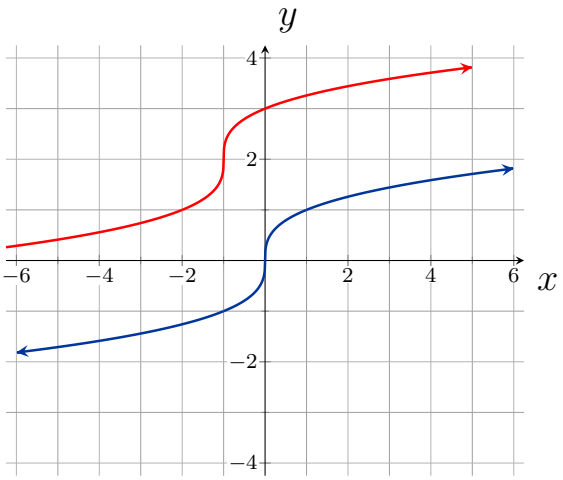
Graph $f(x - 2) + 1$:

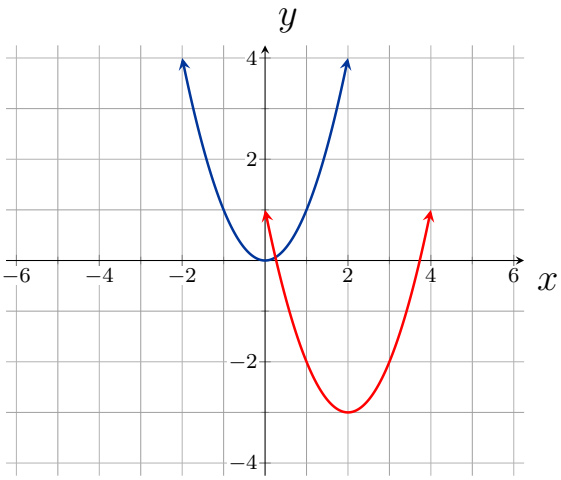


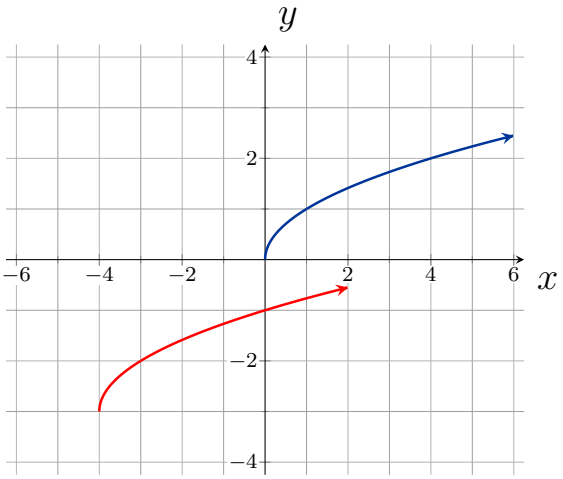
Graph $f(x + 1) - 3$:

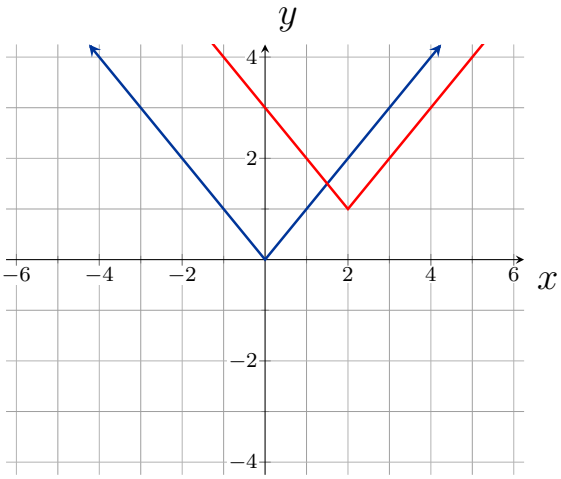


Example. Identify the shifts which transform the blue graph into the red graph:







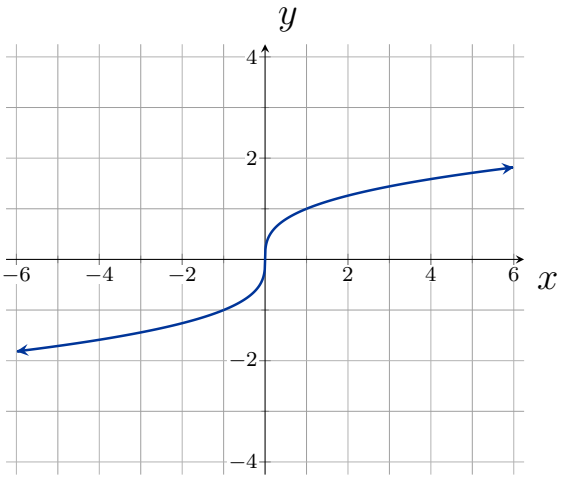


Definition. (Reflections)
Given a function $f(x)$, a new function $g(x) = -f(x)$ is a **vertical reflection** of the function $f(x)$, sometimes called a reflection about the x-axis.

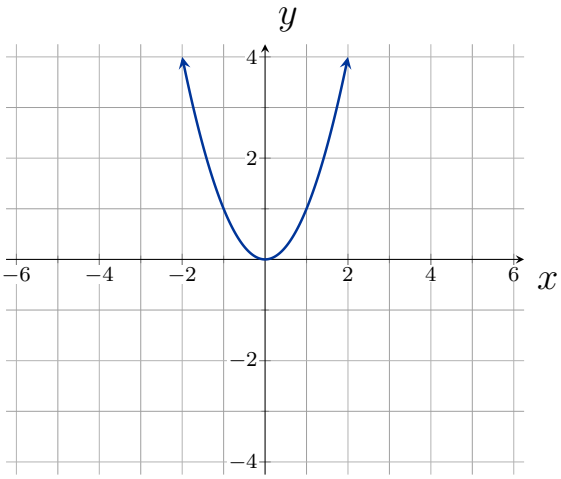
Given a function $f(x)$, a new function $g(x) = f(-x)$ is a **horizontal reflection** of the function $f(x)$, sometimes called a reflection about the y-axis.

Example. Transform the graphs below as indicated:

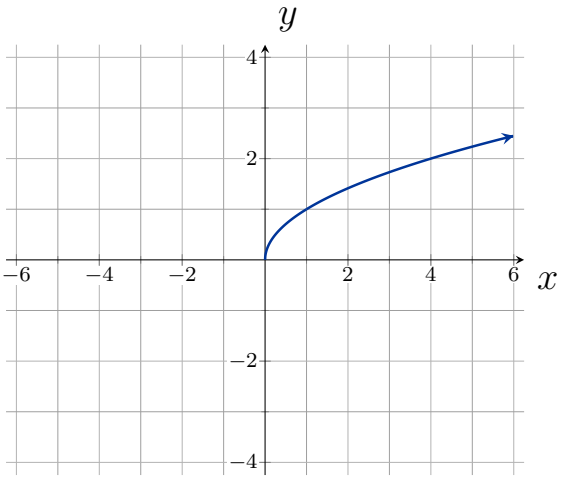
Graph $f(-x)$:



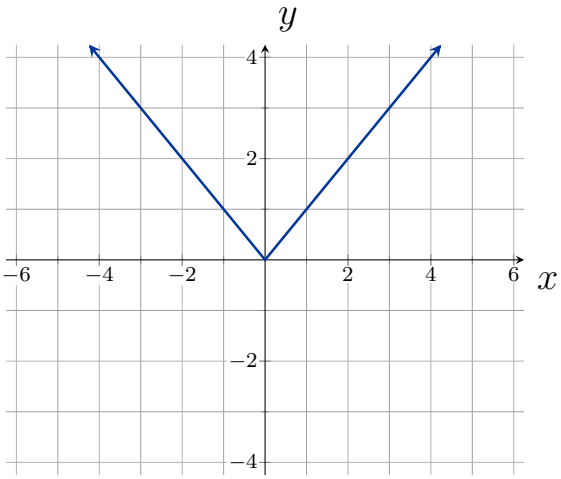
Graph $-f(x)$:



Graph $-f(-x)$:



Graph $-f(x) + 1$



Definition. (Even and Odd functions)
A function is called an even function if for every input x

$$f(x) = f(-x)$$

The graph of an even function is symmetric about the $y-$axis.

A function is called an odd function if for every input x

$$f(x) = -f(-x) \qquad \text{or} \qquad -f(x) = f(-x)$$

The graph of an odd function is symmetric about the origin.

Example. Using algebra, determine if the following functions are even, odd, or neither: Graphs

$f(x) = x^3$

$g(x) = |x| + 1$

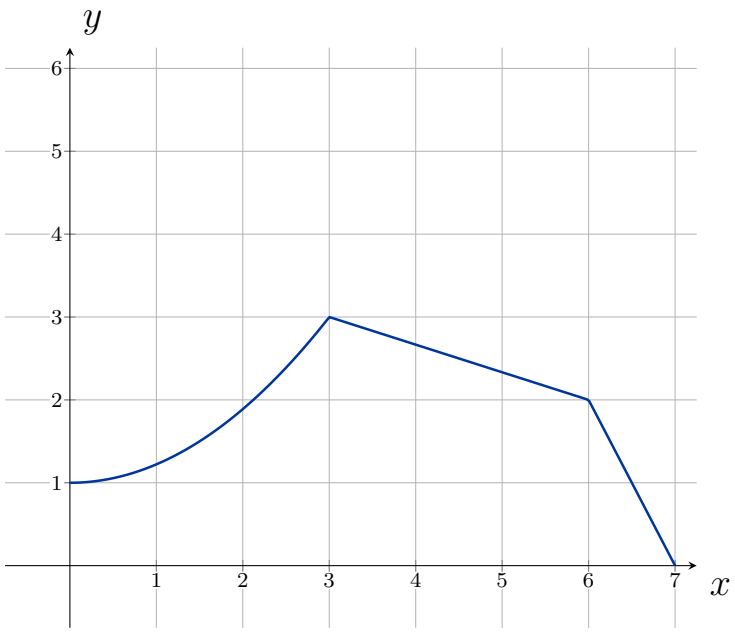
$h(x) = (x-1)^2$

$j(x) = x^3 - \dfrac{1}{x}$

Definition. (Vertical Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = af(x)$, where a is a constant, is a **vertical stretch** or **vertical compression** of the function $f(x)$.

- If $a > 1$, then the graph will be stretched.
- If $0 < a < 1$, then the graph will be compressed.
- If $a < 0$, then there will be a combination of a vertical stretch/compression with a vertical reflection.

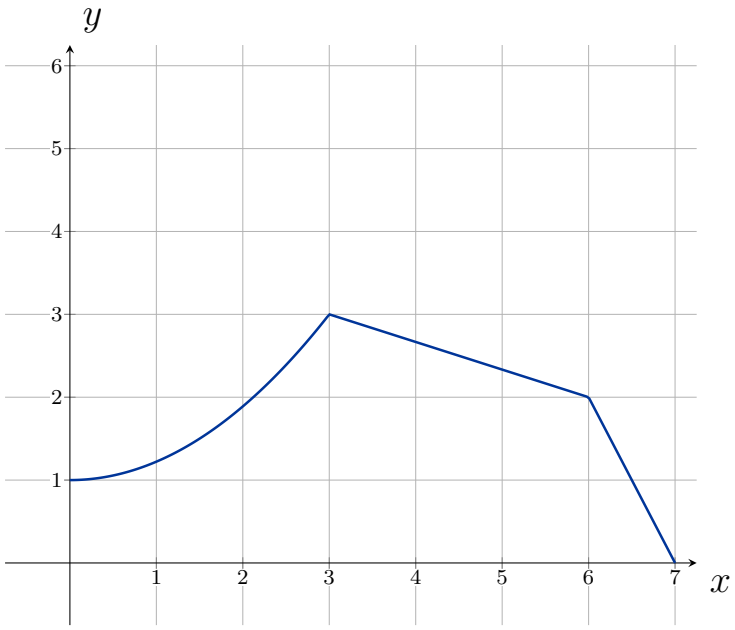
Example. Given the graph of $f(x)$ below, graph $g(x) = 2f(x)$: Graphs



Definition. (Horizontal Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = f(bx)$, where b is a constant, is a **horizontal stretch** or **horizontal compression** of the function $f(x)$.

- If $b > 1$, then the graph will be stretched by $\frac{1}{b}$.
- If $0 < b < 1$, then the graph will be compressed by $\frac{1}{b}$.
- If $b < 0$, then there will be a combination of a horizontal stretch/compression with a horizontal reflection.

Example. Given the graph of $f(x)$ below, graph $g(x) = f(2x)$: Graphs

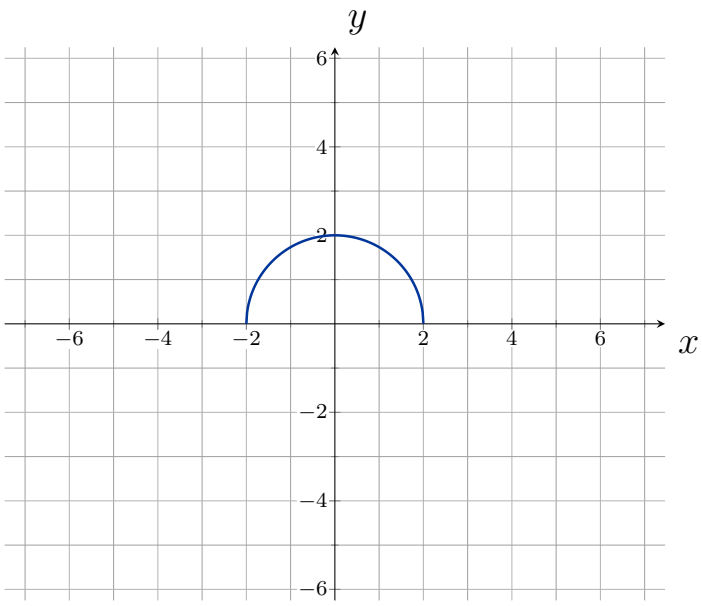


Combining Transformations

When combining transformations ...

- $af(x)+k$: first vertically stretch by a and then shift by k.
- $f(bx-h)$: first horizontally shift by h and then horizontally stretch by $\frac{1}{b}$.
- $f(b(x-h))$: first horizontally stretch by $\frac{1}{b}$ and then horizontally shift by h.
- Horizontal and vertical transformations are independent of the order they are performed.

Example. Given the graph of $f(x)$ below, graph $g(x)=f\left(\frac{1}{2}x+1\right)-3$: Graphs



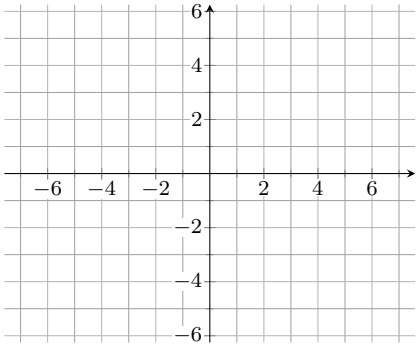
1.6: Absolute Value Functions

Definition. (Absolute Value Function)
The absolute value function can be defined as a piecewise function

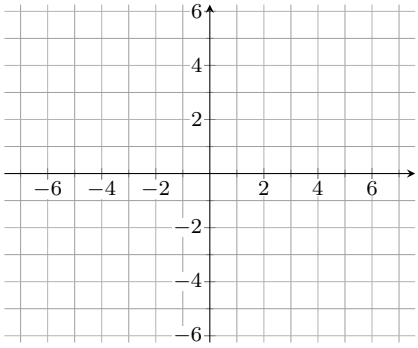
$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example. Graph the following

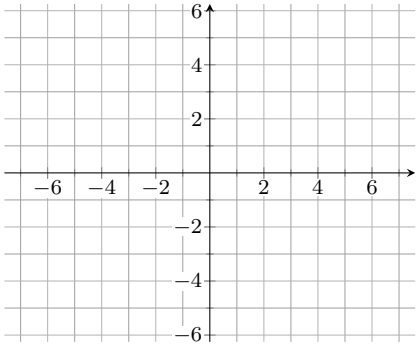
$f(x) = |x|$



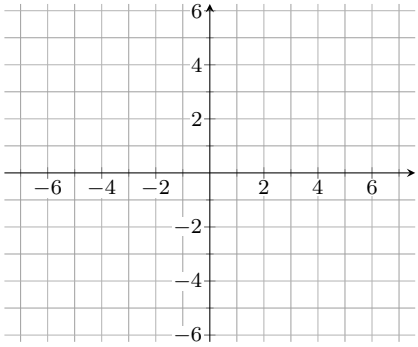
$f(x) = |x - 2|$



$f(x) = |x| + 3$



$f(x) = 2|x - 3| + 1$



Example. Solve the following equations

Graphs

$|x| = 3$

$|2x + 6| = 2$

$|x + 3| - 1 = 0$

$2 - |4 - x| = 0$

Example. Solve the following inequalities. Give your answer in interval notation. Graphs

$|x - 5| \leq 4$

$\left|\frac{x}{4} - 1\right| < 2$

$|2 - x| \leq 3$

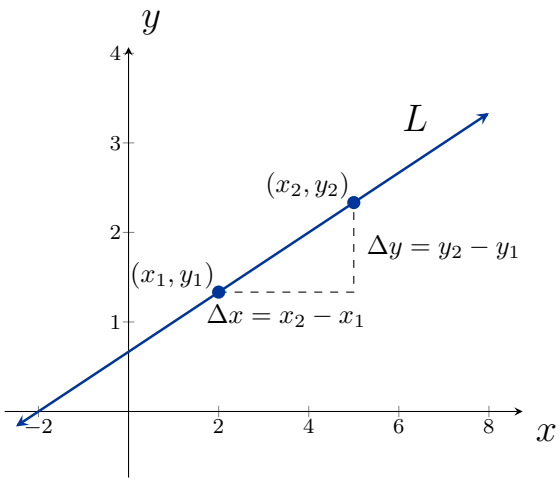
$|x - 3| > 2$

2.1: Linear Functions

Definition. (Slope of a Nonvertical Line)
If (x_1, y_1) and (x_2, y_2) are any two distinct points on a nonvertical line L, then the slope m of L is given by

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example. Compute the slope of the line passing through the points



$(x_1, y_1) = (1, 1)$ and $(x_2, y_2) = (4, 2)$

$(x_1, y_1) = (3, -1)$ and $(x_2, y_2) = (-2, 8)$

$(x_1, y_1) = (3, 2)$ and $(x_2, y_2) = (-1, 2)$

$(x_1, y_1) = (4, 1)$ and $(x_2, y_2) = (4, 4)$

Definition. (Point-Slope Form of an Equation of a Line)
An equation of the line that has slope m and passes through the point (x_1, y_1) is given by

$$y - y_1 = m(x - x_1)$$

Example. Find the equation of the line going through the points

$(x_1, y_1) = (-2, 1)$ and $(x_2, y_2) = (3, -2)$

$(x_1, y_1) = (3, 4)$ and $(x_2, y_2) = (-1, 4)$

$(x_1, y_1) = (2, 0)$ and $(x_2, y_2) = (2, 1)$

Definition. (Slope-Intercept Form of an Equation of a Line)
An equation of the line that has slope m and intersects the y-axis at the point $(0, b)$ is given by

$$y = mx + b$$

Example. Rewrite the equations in the previous example in slope-intercept form.

Example. Marcus currently has 200 songs in his music collection. Every month, he adds 15 new songs. Write a formula for the number of songs, N, in his collection as a function of time, t, the number of months. How many songs will he own in a year?

Forms of Linear Equations

General form:	$Ax + By = C$
Point-slope form:	$y - y_1 = m(x - x_1)$
Slope-intercept form:	$y = mx + b$
Vertical line:	$x = a$
Horizontal line:	$y = b$

2.2: Graphs of Linear Functions

Definition. (Parallel and Perpendicular lines)
Let L_1 and L_2 be lines with slopes m_1 and m_2 respectively. If L_1 and L_2 are *parallel*, then

$$m_1 = m_2.$$

If L_1 and L_2 are *perpendicular*, then

$$m_1 = -\frac{1}{m_2}.$$

Example.

Find the line *parallel* to $y = \frac{3}{2}x + 1$ that passes through the point $(-4, 10)$.

Find the line *perpendicular* to $y = \frac{3}{2}x + 1$ that passes through the point $(-3, 4)$.

Definition.
The point(s) where a graph intersects the axes are called intercepts.
The x**-intercepts** are the points where a function intersects the x-axis.
The y**-intercept** is the point where a function intersects the y-axis.

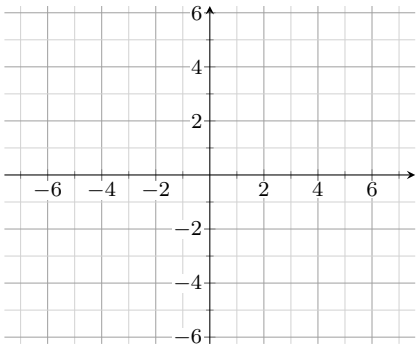
- To solve for the y-intercept:
  - Set $x = 0$,
  - Solve for y.
- To solve for the x-intercept:
  - Set $y = 0$,
  - Solve for x.

Example. Find the x and y intercepts of the following:

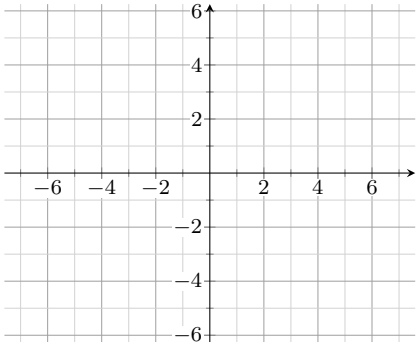
$$y = \frac{4}{5}x - 8 \qquad 3x + 2y = 18$$

Example. Graph the following:

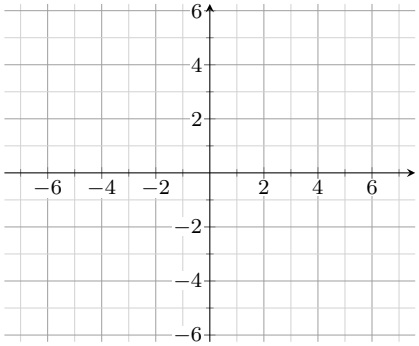
$y = -\frac{3}{2}x + 3$



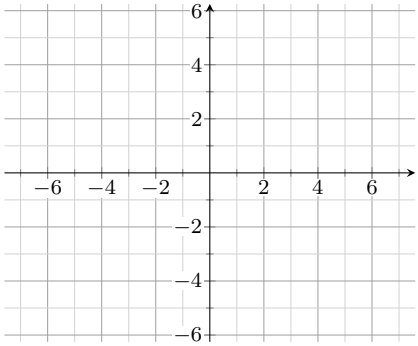
The line through $(-4, 2)$ and $(1, 4)$



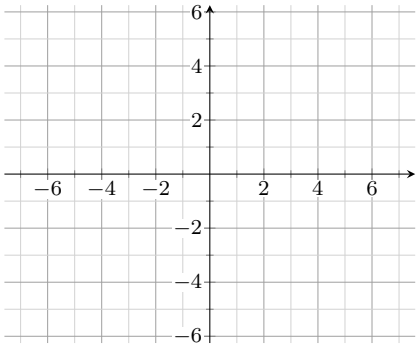
The horizontal line through $(2, 3)$



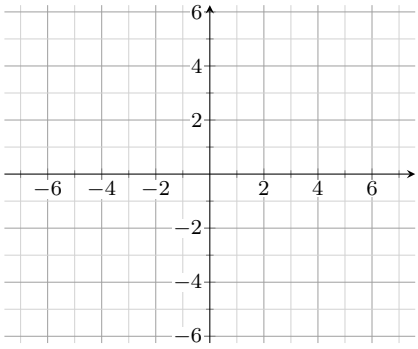
The vertical line through $(-1, 5)$



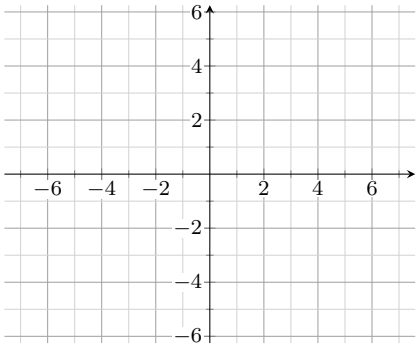
$8x - 10y = 40$



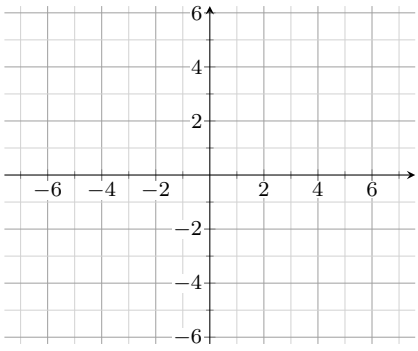
$y = -4x - 5$



Through $(1, 1)$, parallel to $y = 2x - 4$



Through $(3, -2)$, perp. to $y = \frac{1}{3}x - 1$

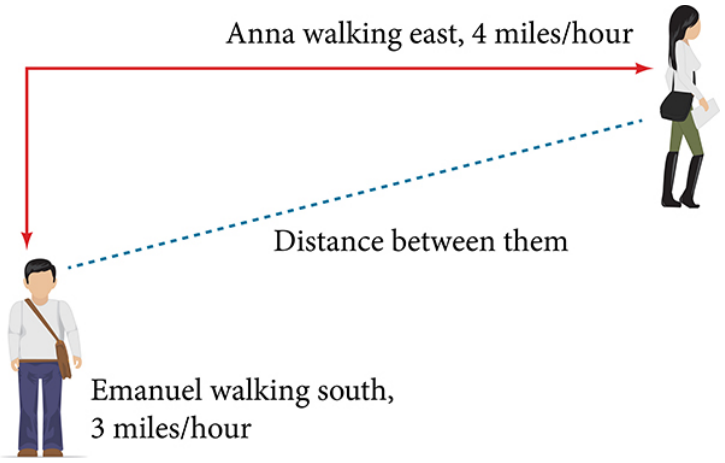


2.3: Modeling with Linear Functions

Example. Elan is a college student who plans to spend a summer in Seattle. Elan has saved $\$3,500$ for the trip and anticipates spending \$400 each week on rent, food, and activities. Write a linear model to represent this scenario. Identify and interpret:

- the rate of change
- the initial amount
- the x-intercept

Example. Anne and Emanuel start at the same intersection. Anna walks east at 4 miles per hour while Emanuel walks south at 3 miles per hour. They are communicating with a two-way radio that has a range of 2 miles. How long after they start walking will they fall out of radio contact?



Example. Jamal is choosing between two truck-rental companies. The first, *Keep on Trucking, Inc.*, charges an up-front fee of \$20, then 59 cents a mile. The second, *Move It Your Way*, charges an up-front fee of \$16, then 63 cents a mile. When will *Keep on Trucking, Inc.* be the better choice for Jamal?

3.1: Complex Numbers

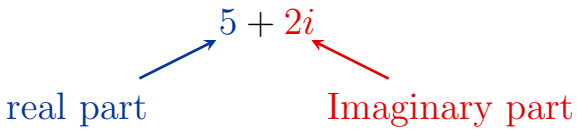
Definition. (Imaginary and Complex Numbers)
We can extend the real numbers by defining:

$$i = \sqrt{-1}.$$

A **complex number** is a number of the form $a + bi$ where

- a is the real part of the complex number, and
- bi is the imaginary part of the complex number.

If $b = 0$, then $a + bi$ is a real number.
If $a = 0$ and $b \neq 0$, then the complex number is called an **imaginary number**.
An imaginary number is an *even* root of a negative number.



Example. Express the following complex numbers in standard form:

$\sqrt{-9}$

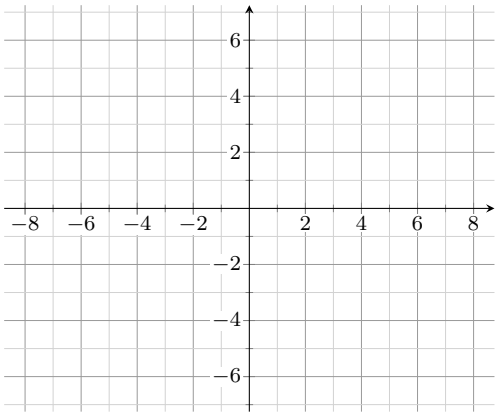
$5 + \sqrt{-3}$

$4 - \sqrt{-16}$

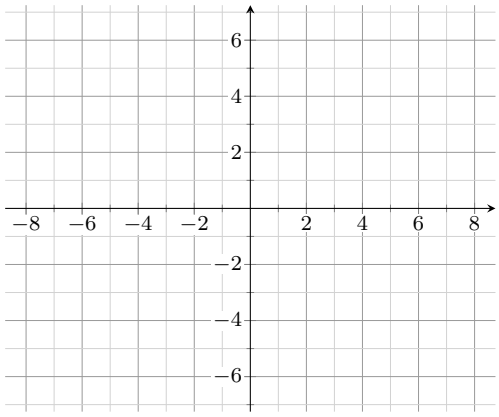
$3 - \sqrt{25}$

Example. Graph the following complex numbers:

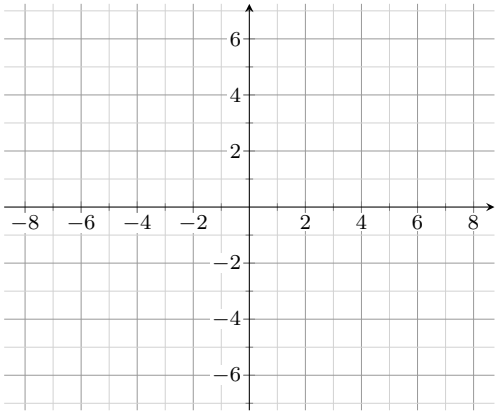
$1 + i$



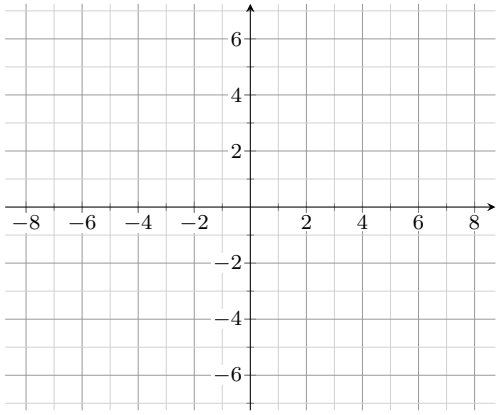
$-2 + 3i$



$3 - 4i$



$-5 - 2i$



Definition. (Addition and Subtraction of complex numbers)
Given complex numbers $a+bi$ and $c+di$,

$$(a+bi)+(c+di)=(a+c)+(b+d)i$$

$$(a+bi)-(c+di)=(a+c)-(b+d)i$$

Example. Evaluate the following:

$(3+2i)+(1-4i)$

$(-4-2i)-(-1+5i)$

$(2)-(-1+2i)$

$(5i)+(3+i)$

Definition. (Multiplying complex numbers)
Given complex numbers $a + bi$ and $c + di$,

$$\begin{aligned}(a + bi) \cdot (c + di) &= ac + adi + bci + bdi^2 \\ &= ac + adi + bci - bd \\ &= (ac - bd) + (ad + bc)i\end{aligned}$$

Example. Evaluate the following:

$(3 + 2i) \cdot (1 - 4i)$

$(-4 - 2i) \cdot (-1 + 5i)$

$(2) \cdot (-1 + 2i)$

$(5i) \cdot (3 + i)$

Definition. (Dividing complex numbers)
Given complex numbers $a+bi$ and $c+di$, division is done via the *complex conjugate*:

$$\begin{aligned}\frac{c+di}{a+bi}\cdot\frac{a-bi}{a-bi} &= \frac{(c+di)(a-bi)}{(a+bi)(a-bi)}\\ &= \frac{ac-bci+adi-bdi^2}{a^2-abi+abi-b^2i^2}\\ &= \frac{(ac+bd)+(ad-bc)i}{a^2+b^2}\end{aligned}$$

Example. Evaluate the following:

$$\frac{3+2i}{1-4i}$$

$$\frac{-4-2i}{-1+5i}$$

$$\frac{2}{-1+2i}$$

$$\frac{3+i}{5i}$$

Example. Given $f(x) = x^2 - 5x + 2$, evaluate $f(3 + i)$.

Example. Given $f(x) = \dfrac{2 + x}{x + 3}$, evaluate $f(10i)$.

Example. Find i raised to different powers: i, i^2, i^3, $\ldots$, then evaluate i^{35}.

3.2: Quadratic functions

Definition.
A quadratic is a function of degree two. The **general form of a quadratic function** is

$$f(x) = ax^2 + bx + c \quad (a \neq 0)$$

where a, b, and c represent constants. The **standard form of a quadratic** is

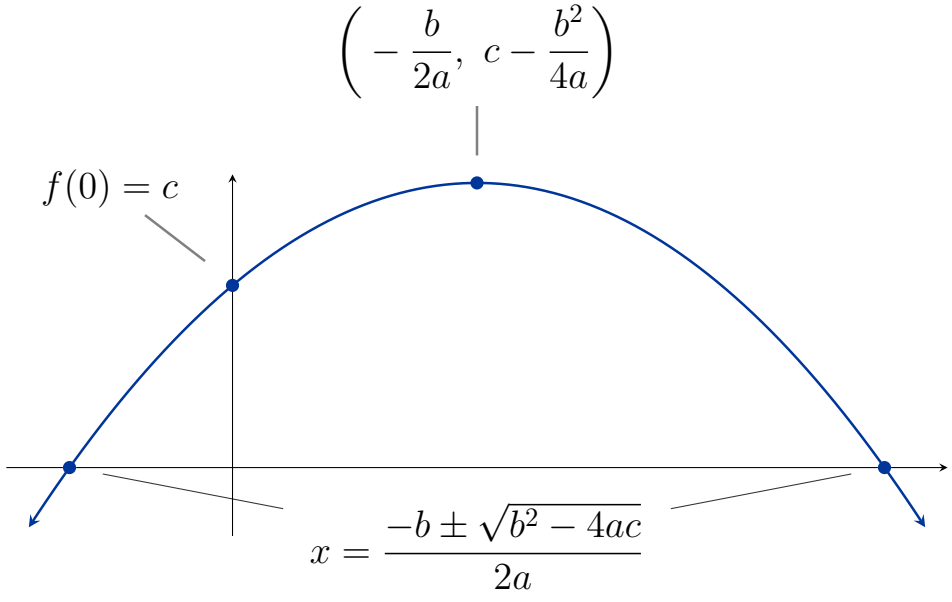
$$f(x) = a(x - h)^2 + k.$$

A **parabola** is the shape of the graph of a quadratic function, with a vertex at (h, k):

$$h = -\frac{b}{2a}, \quad \text{and} \quad k = f(h) = c - \frac{b^2}{4a}$$

The optimal value occurs at the vertex of a parabola:

- A maximum if $a < 0$ 
- A minimum if $a > 0$ 



Example. Expand the standard form of a quadratic, then write h and k in terms of a, b, and c.

Example. Find the vertex of $f(x) = 2x^2 - 6x + 7$ by

- computing h and k, and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

- Factor "a" from the first two terms

- Add and subtract h^2

- Bring $-h^2$ out

- Simplify

Example. Find the vertex of $g(x) = 13 - 3x^2 + 4x$ by

- computing h and k, and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

- Factor "a" from the first two terms

- Add and subtract h^2

- Bring $-h^2$ out

- Simplify

Example. For the following quadratic functions, find

- the x and y-intercepts,
- the vertex, and
- the domain and range.

$f(x) = 3x^2 + 2x - 4$

$g(x) = -(x+2)^2 - 1$

Solving quadratic equations via factoring:

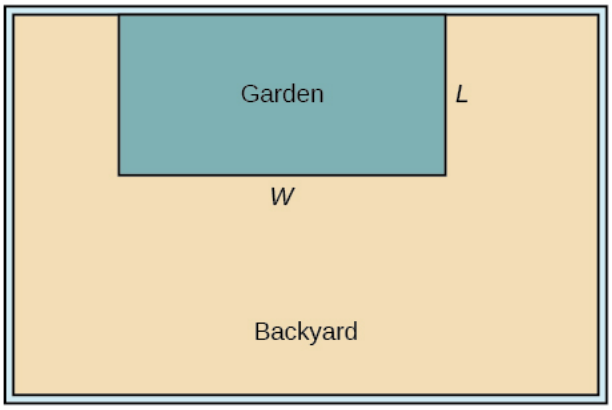
Example. Solve $2x^2 + x = 3x + 12$

1. Rewrite the equation in the general form: $2x^2 - 2x - 12 = 0$
2. Rewrite bx using factors of ac: $2x^2 - 6x + 4x - 12 = 0$
3. Factor out like terms: $2x(x-3) + 4(x-3) = 0$
4. Factor by grouping: $(x-3)(2x+4) = 0$
5. Solve for the roots: $x = 3$ and $x = -2$

Example. Solve the following for x via factoring:

$(x+3)(x-1) = 5$ $\qquad$ $-4x^2 + 8x - 3 = 0$

Example. A backyard farmer wants to enclose a rectangular space for a new garden within her fenced backyard. She has purchased 80 feet of wire fencing to enclose three sides, and she will use a section of the backyard fence as the fourth side (see below). What dimensions should she use to maximize the enclosed area?



3.3: Power Functions and Polynomial Functions

Definition. (Power Function)
A **power function** is a function that can be represented in the form

$$f(x) = kx^p$$

where k and p are real numbers, and k is known as the **coefficient**.

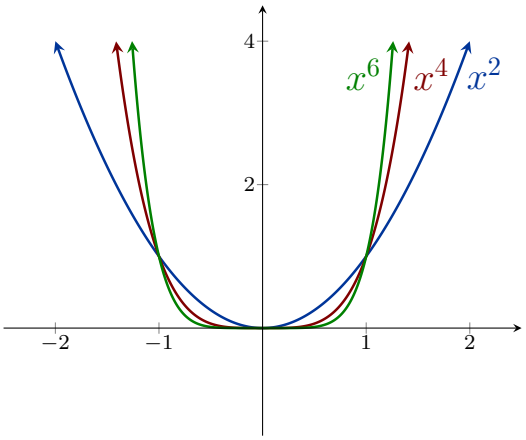
Example. Identify which of the following functions are power functions:

$f(x) = 1$	Constant	$f(x) = x$	Identity
$f(x) = x^2$	Quadratic	$f(x) = x^3$	Cubic
$f(x) = \dfrac{1}{x}$	Reciprocal	$f(x) = \dfrac{1}{x^2}$	Reciprocal squared
$f(x) = \sqrt{x}$	Square root	$f(x) = \sqrt[3]{x}$	Cube root
$f(x) = 2^x$	Exponential		

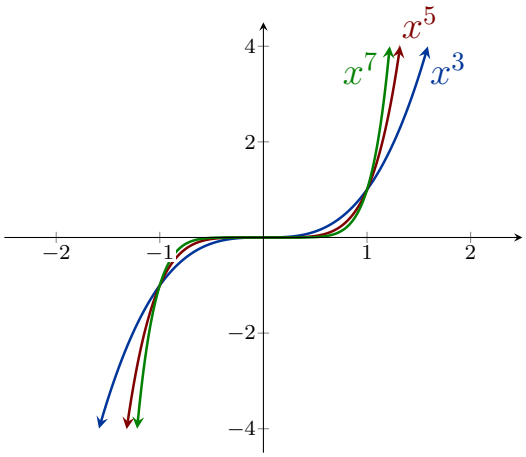
Identifying End Behavior of Power Functions

Below are example graphs of power functions $f(x) = kx^n$:

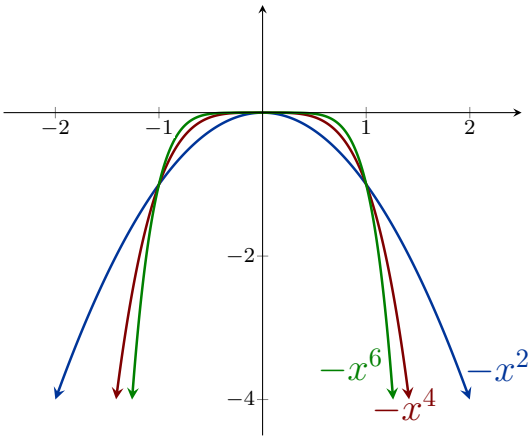
n Even, $k > 0$



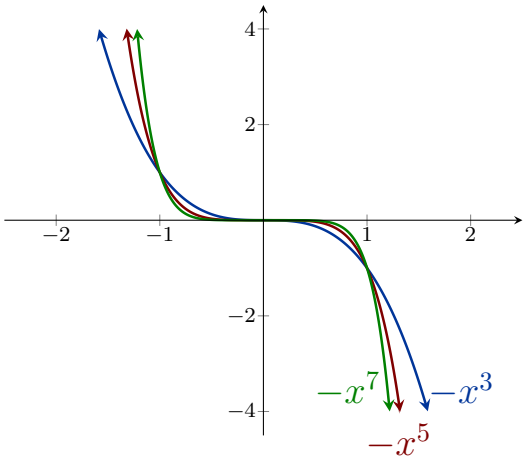
n Odd, $k > 0$



n Even, $k < 0$



n Odd, $k < 0$



Example. Describe the end behavior of the graphs of the following power functions:

$f(x) = x^8$ $\qquad\qquad$ $g(x) = -x^9$

Definition. (Polynomial Functions)
Let n be a non-negative integer. A **polynomial function** is a function that can be written in the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

Each a_i is a coefficient and can be any real number, and $a_n \neq 0$.

The **degree** of the polynomial is the highest power that the variable is raised to. The **leading coefficient** is the coefficient associated with the term containing the highest power of the variable.

Example. Describe the end behavior of the graphs of the following polynomials: Graph

$f(x) = 5x^4 + 2x^3 - x - 4$ $\qquad\qquad$ $g(x) = -2x^6 - x^5 + 3x^4 + x^3$

$h(x) = 3x^5 - 4x^4 + 2x^2 + 1$ $\qquad\qquad$ $j(x) = -6x^3 + 7x^2 + 3x + 1$

Definition. (Intercepts and Turning Points of Polynomial Functions)
A **turning point** is a point at which the graph changes direction from increasing to decreasing, or decreasing to increasing.

The y-intercept is the point at which the function has an input value of zero.

The x-intercepts are the points at which the output value is zero.

A polynomial of degree n will have, at most, n x-intercepts, and $n-1$ turning points.

Complex Conjugate Root Theorem
If a polynomial $f(x)$ has real coefficients, and $a+bi$ is a root of $f(x)$, then $a-bi$ is also a root of $f(x)$.

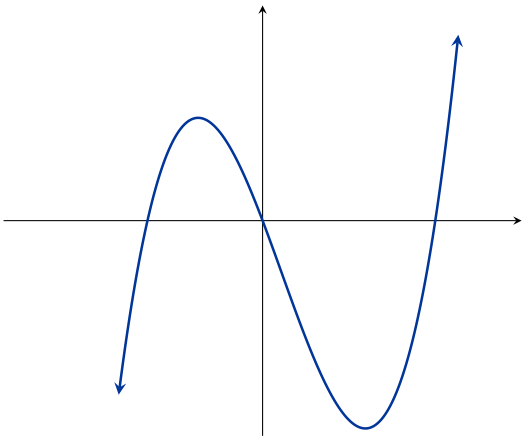
Example. Find the intercepts of

$$f(x) = x^4 - 4x^2 - 45$$

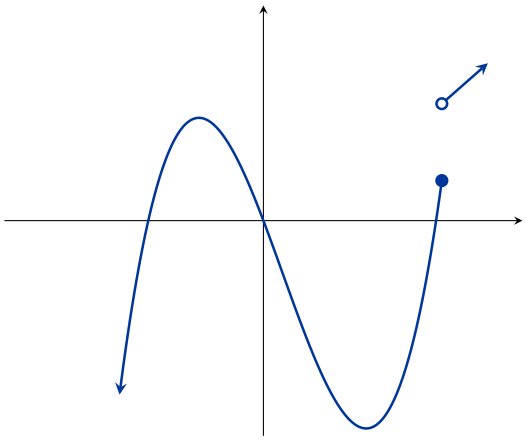
3.4: Graphs of Polynomial Functions

Graphs of polynomials are smooth and continuous with infinite end behavior.

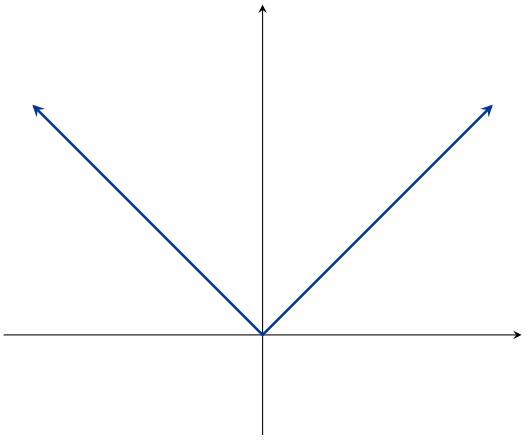
Smooth and continuous



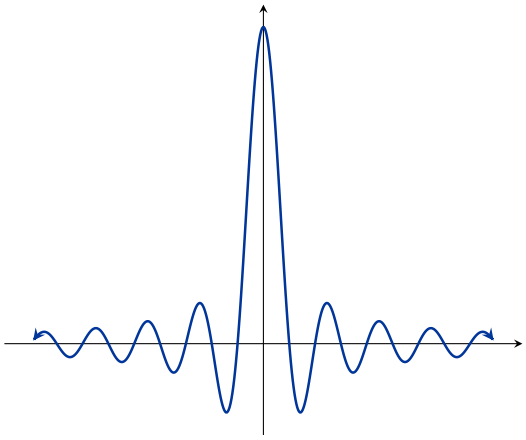
Not continuous



Not smooth



Finite end behavior



Example. Find the x- and y-intercepts of the following functions:

$f(x) = 2(x-4)(x+1)(x-6)$ $\qquad$ $g(x) = 4x(x-2)^2(x+1)$

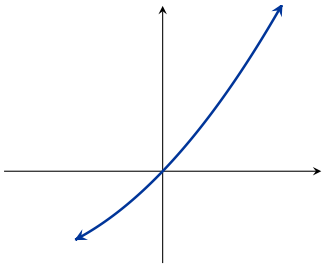
$h(x) = x^4 - x^2$ $\qquad$ $j(x) = x^3 + 2x^2 - 9x - 18$

Definition. (Multiplicity of a Root)
If a polynomial contains a factor of the form $(x-h)^p$, then the behavior near the x-intercept h is determined by the power p. We say that $x=h$ is a zero of **multiplicity** p.
When p is,

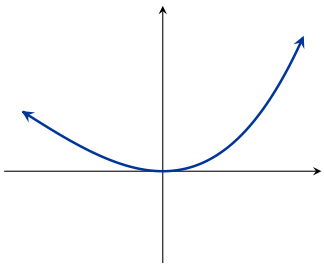
- even, the graph will touch the x-axis
- odd, the graph will cross the x-axis

The degree of the polynomial is the sum of the multiplicities.

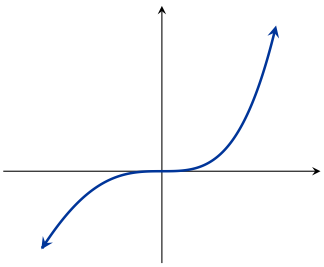
Single zero



Zero with multiplicity 2



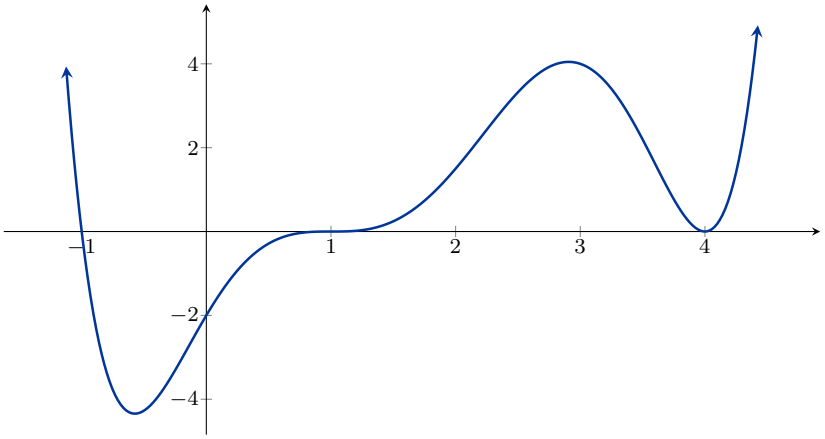
Zero with multiplicity 3



Example. For the following, find the zeros, and the multiplicities of each:

$f(x) = (x+2)^3(x-3)^2(x^2-1)$ $\qquad$ $g(x) = x^5(x-1)^3(x+2)$

Example. Find a formula for the polynomial of minimum degree represented in the graph below:



3.5: Dividing Polynomials

Example. Compute $178 \div 3$ using long division. Identify the dividend, divisor, quotient, and remainder.

Definition. (The Division Algorithm)
The **Division Algorithm** states that, given a polynomial dividend $f(x)$ and a non-zero polynomial divisor $d(x)$ where the degree of $d(x)$ is less than or equal to the degree of $f(x)$, there exist unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)q(x) + r(x)$$

where $q(x)$ is the quotient and $r(x)$ is the remainder. The remainder is either equal to zero, or has degree strictly less than $d(x)$.
If $r(x) = 0$, then $d(x)$ divides evenly into $f(x)$. This means that, in this case, both $d(x)$ and $q(x)$ are factors of $f(x)$.

Example. Use polynomial long division on the following:

$$\frac{5x^2 + 3x - 2}{x + 1}$$

$$\frac{2x^2 + 5x - 7}{3x - 2}$$

Definition. (Synthetic Division)
Synthetic division is a shortcut that can be used when the divisor is of the form $(x-k)$. In **synthetic division**, only the coefficients are used in the division process.

Given two polynomials, use synthetic division to divide:

1. Write k for the divisor.
2. Write the coefficients of the dividend.
3. Bring the lead coefficient down.
4. Multiply the lead coefficient by k. Write the product in the next column.
5. Add the terms of the second column.
6. Multiply the result by k. Write the product in the next column.
7. Repeat steps 5. and 6. for the remaining columns.
8. Use the bottom numbers to write the quotient. The number in the last column is the remainder.

Example. Use synthetic division on the following:

$$\frac{5x^2+3x-2}{x+1}$$

-1 | 5 $\quad$ 3 $\quad$ -2

$\quad$ |

$\quad$ |

$$\frac{2x^2+5x-7}{3x-2}$$

$\frac{2}{3}$ | 2 $\quad$ 5 $\quad$ -7

$\quad$ |

$\quad$ |

Example. Use synthetic division on the following:

$$\frac{5x^2 - 3x - 36}{x - 3}$$

$$3 \;\big|\; 5 \quad -3 \quad -36$$

$$\frac{4x^3 + 10x^2 - 6x - 20}{x + 2}$$

$$-2 \;\big|\; 4 \quad 10 \quad -6 \quad -20$$

$$\frac{-9x^4 + 10x^3 + 7x^2 - 6}{x - 1}$$

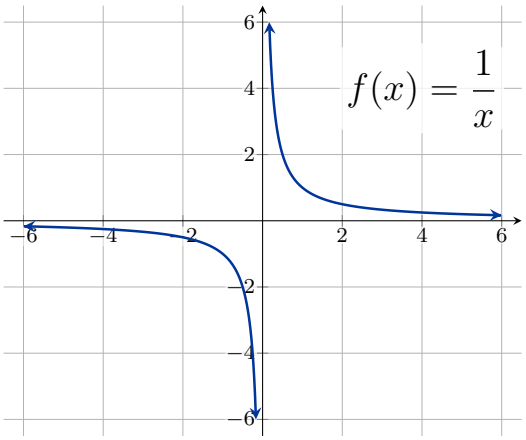
$$1 \;\big|\; -9 \quad 10 \quad 7 \quad 0 \quad -6$$

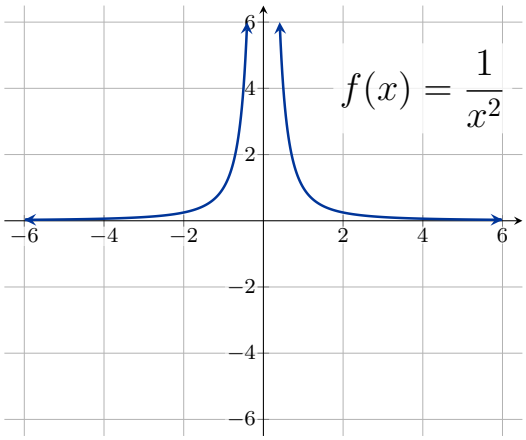
3.7: Rational Functions

Definition. (Rational Function)
A **rational function** is a function of the form

$$r(x) = \frac{f(x)}{g(x)},$$

where $f(x)$ and $g(x)$ are polynomials.





Definition. (Asymptotes)
A **vertical asymptote** of a graph is a vertical line $x = a$ where the graph tends towards positive or negative infinity as the input approaches a from either the left or the write. We write

$$\text{As } x \to a^-,\ f(x) \to \pm\infty \quad \text{or} \quad x \to a^+,\ f(x) \to \pm\infty.$$

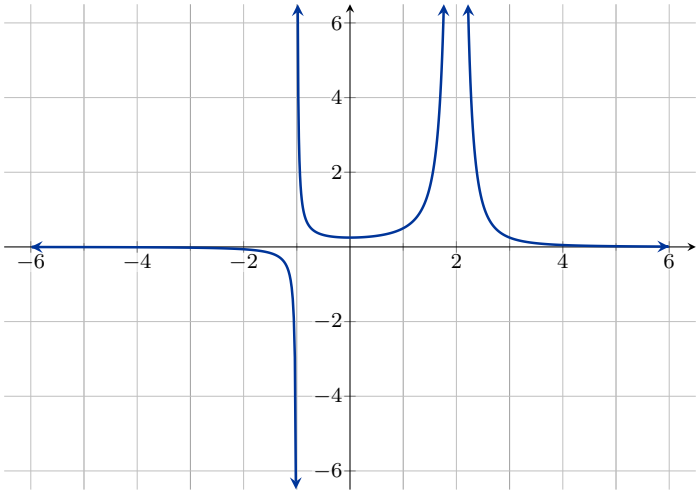
A **horizontal asymptote** of a graph is a horizontal line $y = b$ where the graph approaches the line as the inputs increase or decrease without bound. We write

$$\text{As } x \to -\infty \text{ or } x \to \infty,\ f(x) \to b.$$

Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{1}{(x+1)(x-2)^2}$$

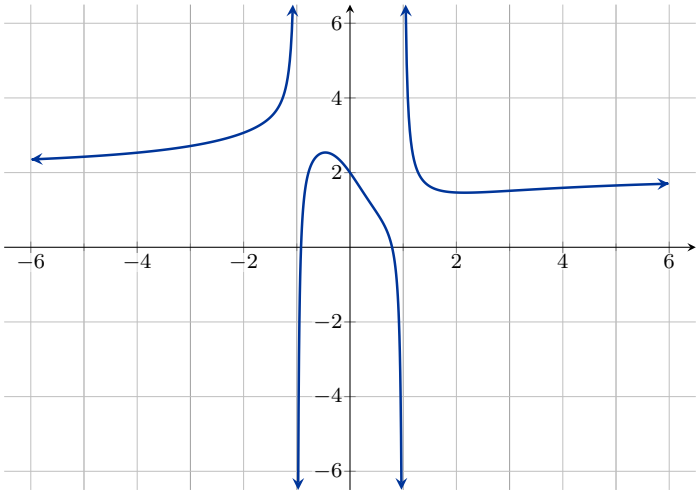
using the graph below:



Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{2x^4 - 2x^3 + x^2 + 2x - 2}{(x^2 - 1)(x^2 + 1)}$$

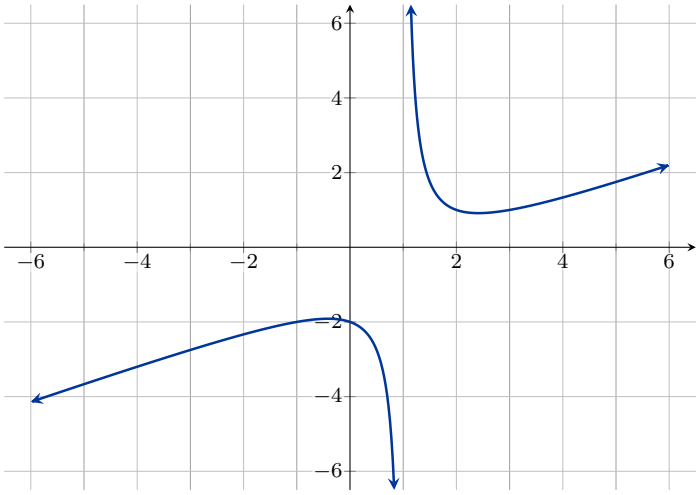
using the graph below:



Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{x^2 - 3x + 4}{2(x-1)}$$

using the graph below:

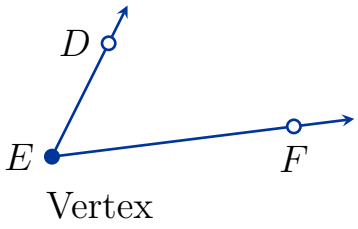


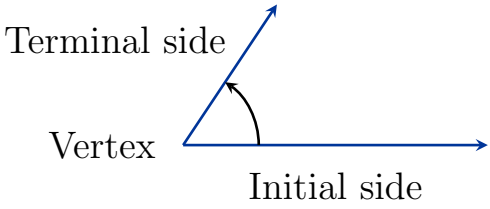
5.1: Angles

Definition.
A **ray** consists of one point on a line, and all points extending in one direction from that point. The following is ray $\overrightarrow{EF}$:

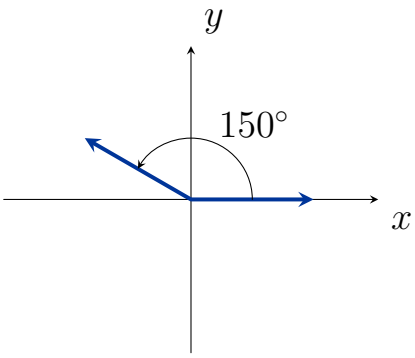


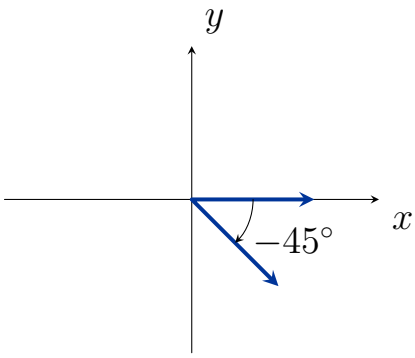
An **angle** is the union of two rays having a common endpoint. The endpoint is called the **vertex** of the angle. Angles can be named using a point on each ray and the vertex. The following is $\angle DEF$:





When we create an angle, we fix one ray, called the **initial side**, and rotate the **terminal side** about the vertex.
An angle drawn on the xy-plane is said to be in **standard position** if its vertex is located at the origin, and the initial side extends along the x-axis:

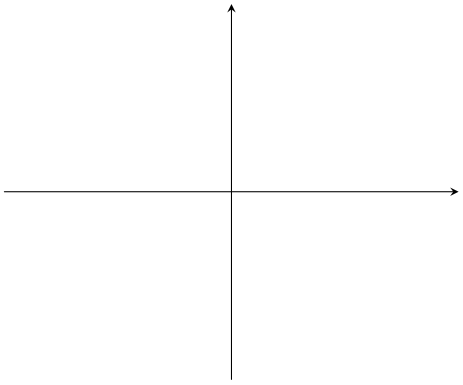




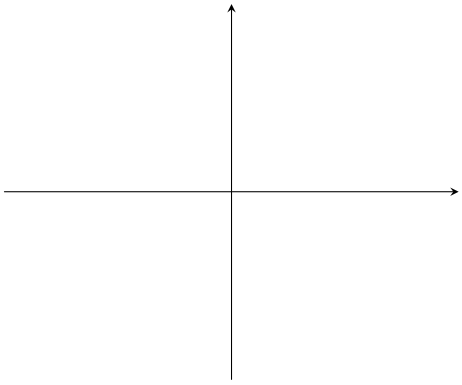
A **positive angle** rotates in the counterclockwise direction. A **negative angle** rotates in the clockwise direction. A full circle rotation is defined to be 360°.

Example. Sketch the following angles in standard position:

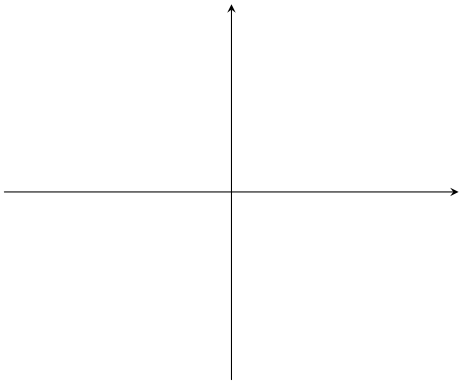
$0°$



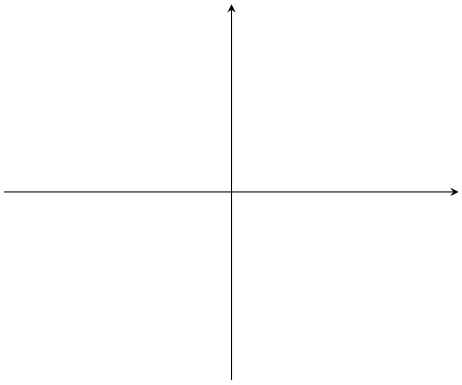
$30°$



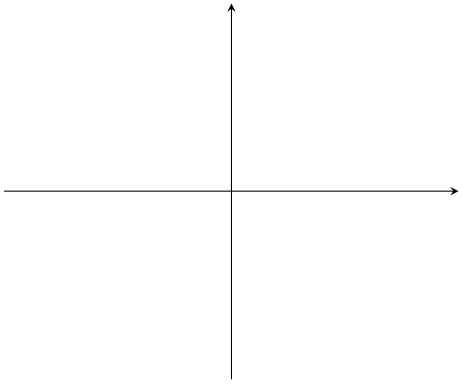
$90°$



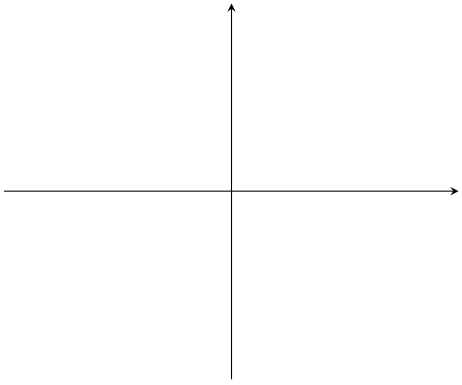
$45°$



$135°$



$-60°$



Definition.
A **radian** is the measure of a central angle corresponding to an arc equal in length to the radius of that circle. Since the circumference of a circle is $C = 2\pi r$, there are 2π radians in a full circle. This gives us

$$2\pi = 360^\circ$$

or, equivalently,

$$\pi = 180^\circ$$

Example. Convert the following angles to radians:

225°

270°

-135°

-45°

Example. Convert the following angles to degrees:

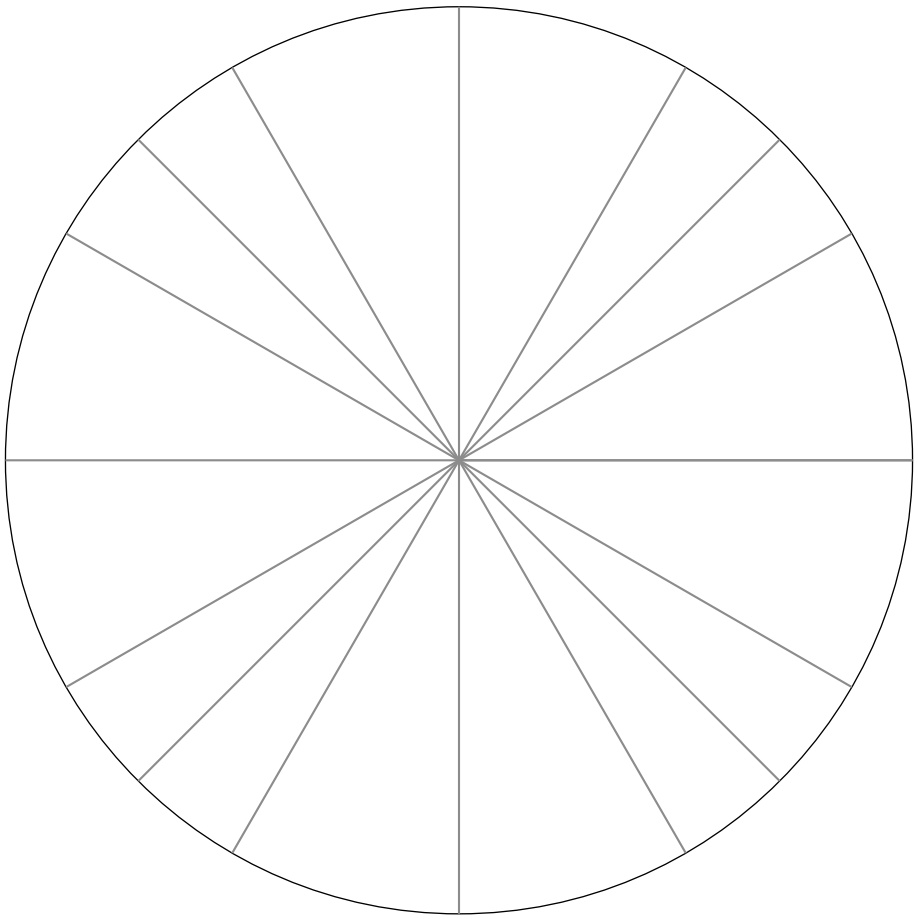
4π

$-\frac{\pi}{2}$

$\frac{\pi}{12}$

$-\frac{5\pi}{4}$

Example. The following circle contains certain angles of interest. Find the angles of each in both degrees and radians:



Definition.
Coterminal angles are two angles in standard position that have the same terminal side.

Example. Find a coterminal angle for each of the following:

400°

-30°

1655°

-500°

7π

$\dfrac{-5\pi}{6}$

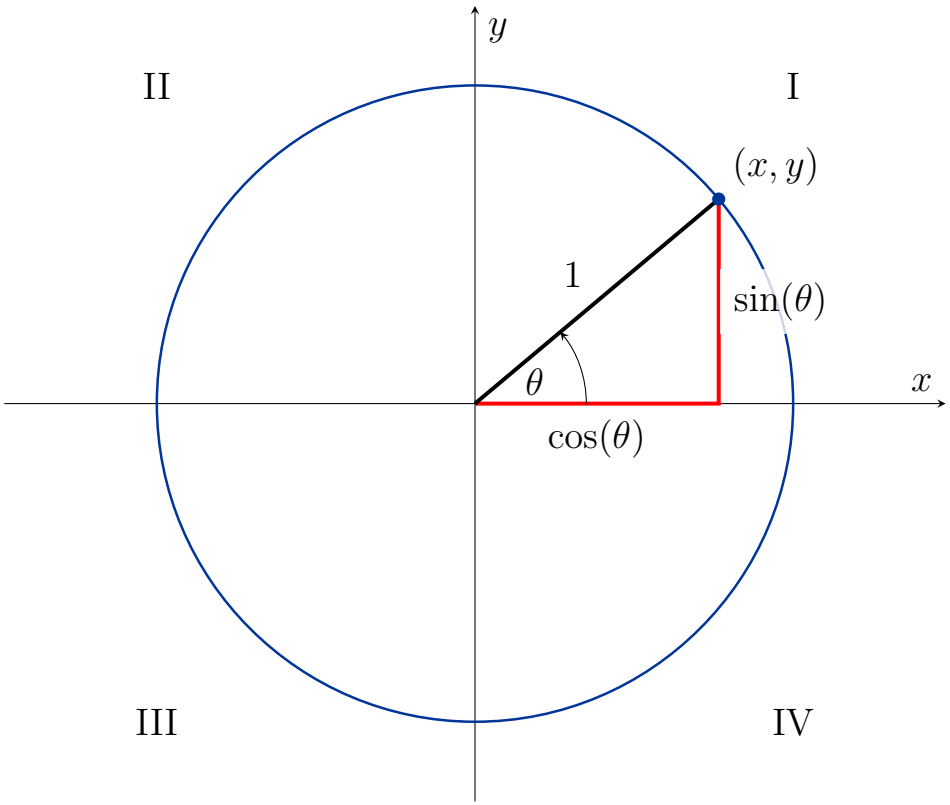
$\dfrac{21\pi}{4}$

$-\dfrac{17\pi}{3}$

5.2: Unit Circle: Sine and Cosine Functions

Definition. (Unit Circle)
A **unit circle** has a center at $(0, 0)$ and radius 1. In a unit circle, the length of the intercepted arc is equal to the radian measure of the central angle θ.

Let (x, y) be the endpoint on the unit circle of an arc of arc length s.
The (x, y) coordinates of this point can be described as functions of the angle.



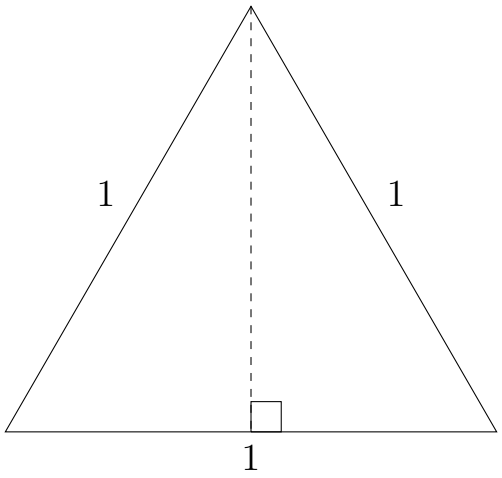
Definition. (Pythagorean Identity)
The **Pythagorean Identity** states that, for any real number θ,

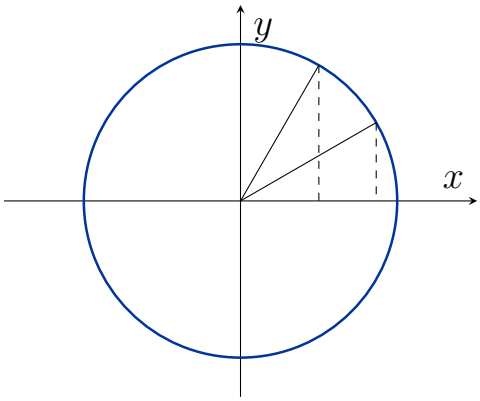
$$\cos^2(\theta) + \sin^2(\theta) = 1$$

Example. If $\sin(\theta) = \frac{3}{7}$ and θ is in the second quadrant, find $\cos(\theta)$.

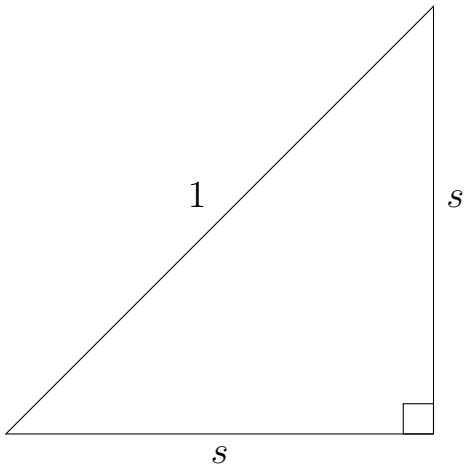
If $\cos(\theta) = \frac{24}{25}$ and θ is in the fourth quadrant, find $\sin(\theta)$.

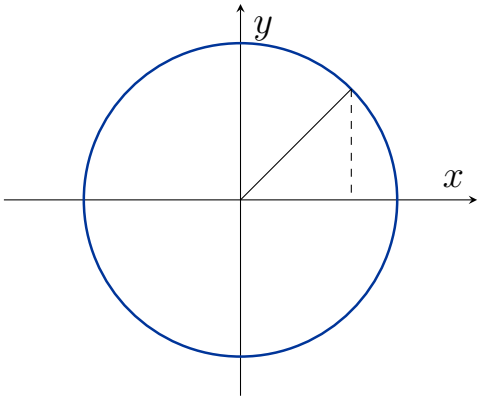
We can find certain values of $\cos(\theta)$ and $\sin(\theta)$ using special triangles:



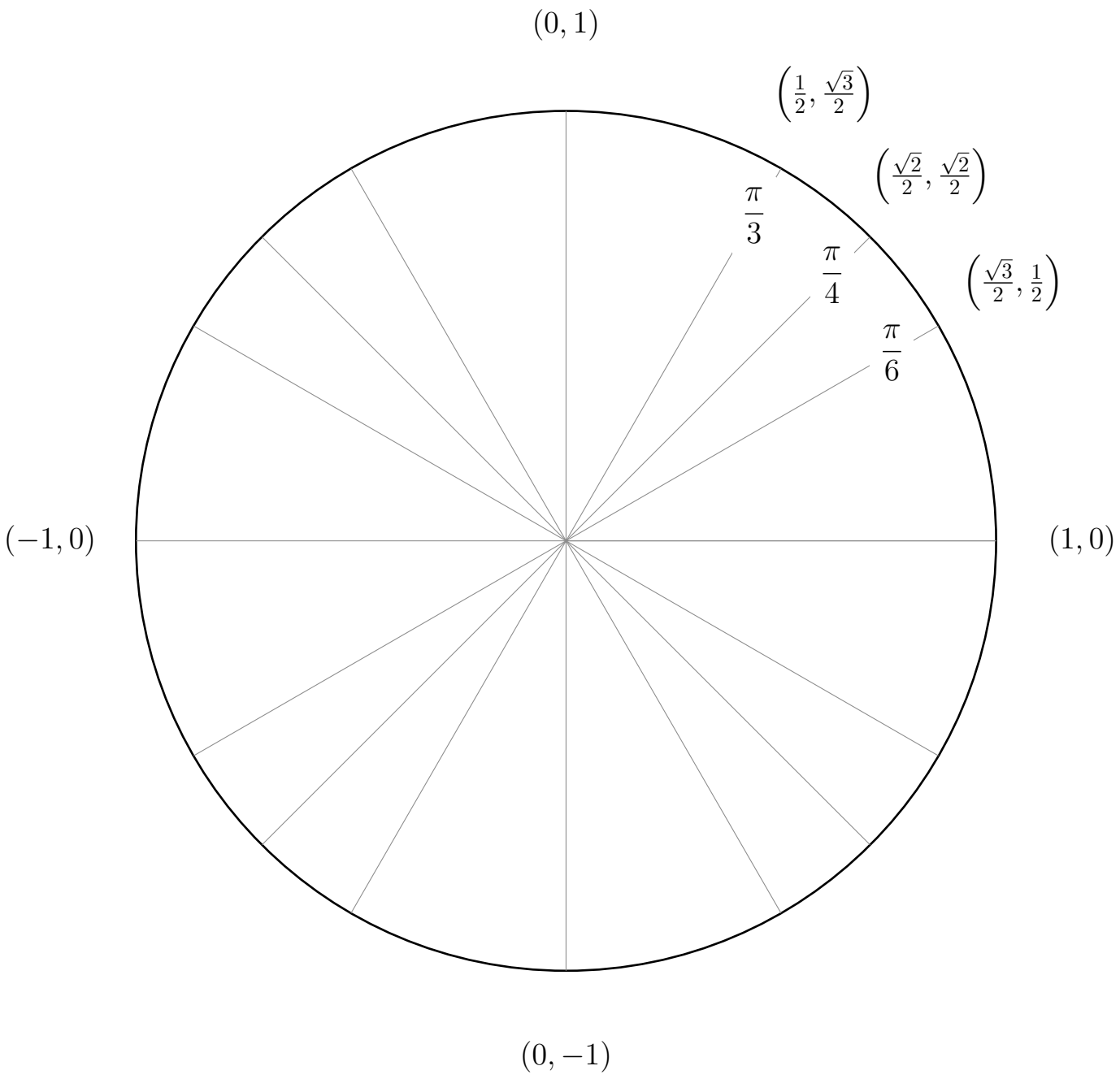


We can find certain values of $\cos(\theta)$ and $\sin(\theta)$ using special triangles:





Example. Fill out the remaining coordinates and radian measures below:



Example. Determine if $\cos(\theta)$ and $\sin(\theta)$ are even, odd, or neither.

Example. State the domain and range of $\cos(\theta)$ and $\sin(\theta)$

Example. Find the coordinate of the point on a circle with radius 15 corresponding to an angle of $220°$.

Example. Find the coordinate of the point on a circle with radius 5 corresponding to an angle of $\frac{3\pi}{4}$.

5.3: The Other Trigonometric Functions

Definition.

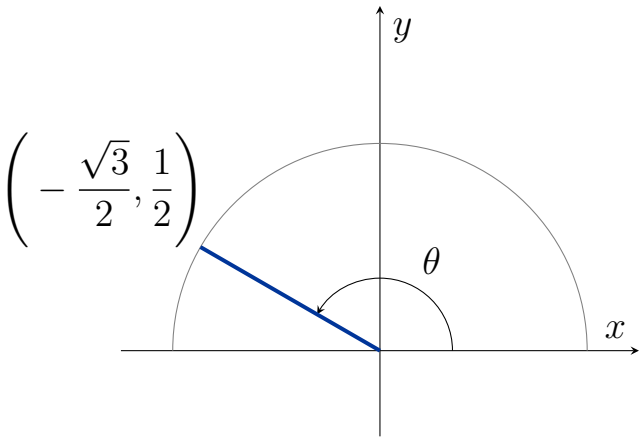
$$\sec(\theta) = \frac{1}{\cos(\theta)} \qquad \tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} \qquad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Example. Simplify the following:

$$\frac{\sec(\theta)}{\tan(\theta)} \qquad \tan(\theta)\cot(\theta)$$

Example. Consider the point $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



$\cos(\theta)$

$\sin(\theta)$

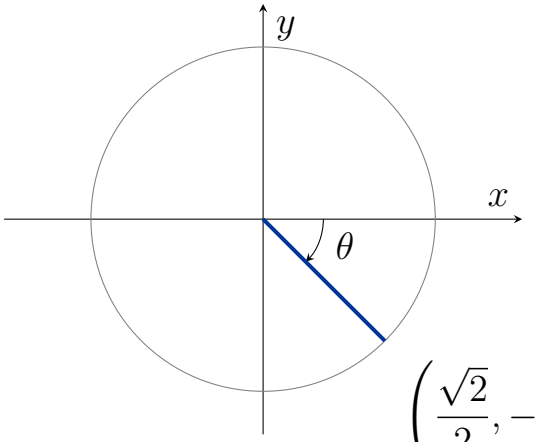
$\sec(\theta)$

$\csc(\theta)$

$\tan(\theta)$

$\cot(\theta)$

Example. Consider the point $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ on the unit circle. Find the value of the six trigonometric functions:



$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

$\cos(\theta)$

$\sin(\theta)$

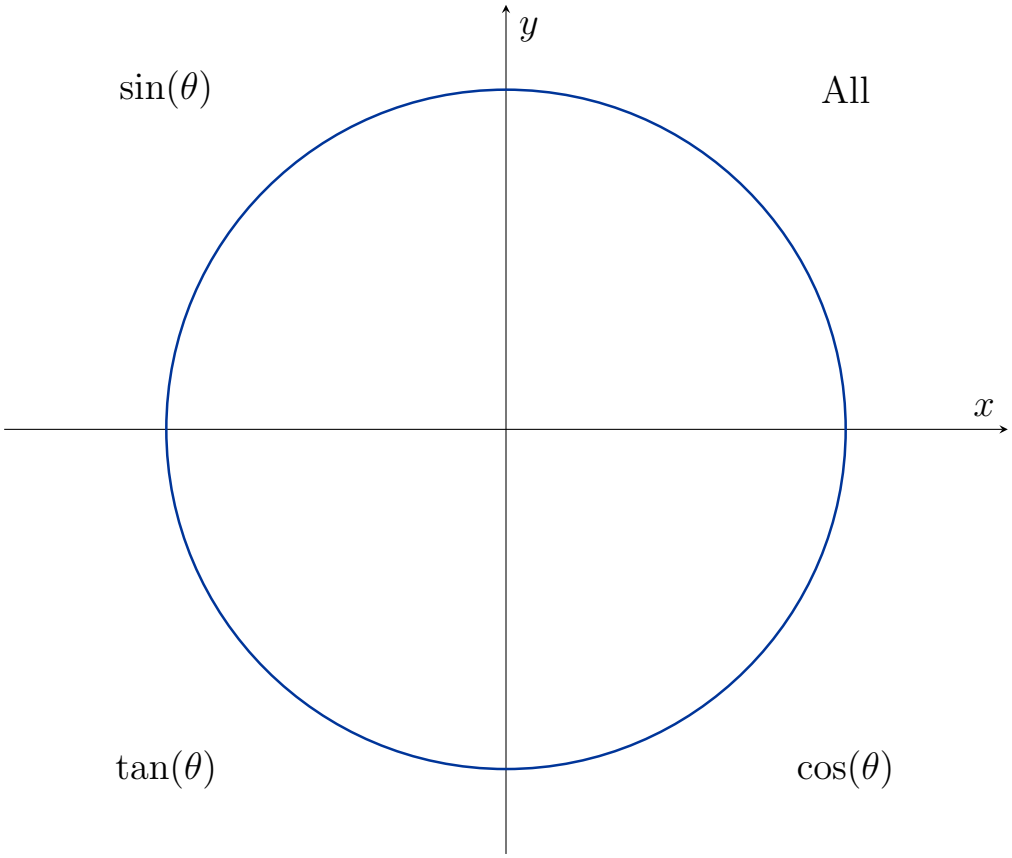
$\sec(\theta)$

$\csc(\theta)$

$\tan(\theta)$

$\cot(\theta)$

Note: The sign of each trig. function is determined by the quadrant



Example. Find the value of the six trigonometric functions when

$\theta = \dfrac{\pi}{6}$ $\theta = \dfrac{\pi}{3}$

$\theta = -\dfrac{5\pi}{6}$ $\theta = -\dfrac{7\pi}{4}$

Recall that $\cos(\theta)$ is even, and $\sin(\theta)$ is odd:

$$\cos(\theta) = \cos(-\theta) \qquad -\sin(\theta) = \sin(-\theta)$$

Thus,

$$\sec(\theta) = \sec(-\theta) \qquad -\csc(\theta) = \csc(-\theta)$$

$$-\tan(\theta) = \tan(-\theta) \qquad -\cot(\theta) = \cot(-\theta)$$

Example. If $\sec(\theta) = 2$, what is $\sec(-\theta)$?

Example. If $\cot(\theta) = \sqrt{3}$, what is $\cot(-\theta)$?

Alternate forms of the Pythagorean Theorem
Recall the Pythagorean Identity:

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

We now define alternate forms:

$$1 + \tan^2(\theta) = \sec^2(\theta)$$
$$\cot^2(\theta) + 1 = \csc^2(\theta)$$

Example. If $\tan(\theta) = -\sqrt{3}$, and $\pi < \theta < 2\pi$, find $\sec(\theta)$.

Example. If $\csc(\theta) = -\dfrac{2\sqrt{3}}{3}$, and $\pi < \theta < \dfrac{3\pi}{2}$, find $\cot(\theta)$.

Definition. (Period of a Function)
The **period** P of a repeating function f is the number representing the interval such that $f(x+P) = f(x)$ for any value of x.

The period for cosine, sine, secant, and cosecant is 2π.

The period for tangent and cotangent is π.

Example. If $\sin(\theta) = \frac{1}{5}$, find $\sin(\theta + 2\pi)$ and $\sin(\theta - 2\pi)$.

Example. If $\tan(\theta) = \frac{\sqrt{3}}{3}$, find $\tan(\theta + \pi)$ and $\cot(\theta + \pi)$.

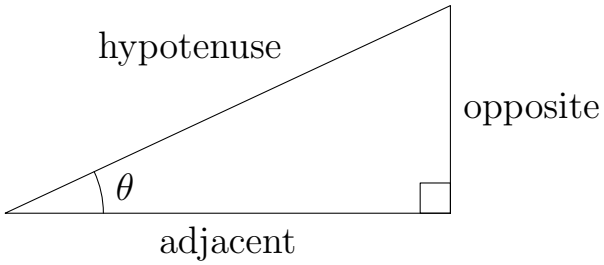
Example. Find the value of all six trigonometric functions and identify which quadrant θ is in:

$$\cos(\theta) = \frac{5}{13} \text{ and } \sin(\theta) = -\frac{12}{13}$$

$$\csc(\theta) = -\sqrt{2} \text{ and } \tan(\theta) = -1$$

5.4: Right Angle Trigonometry

Definition.
Given a right triangle with an acute angle θ



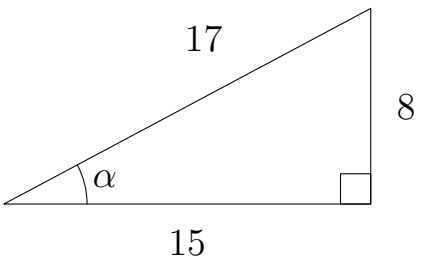
we have

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

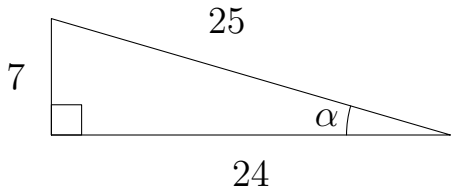
$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

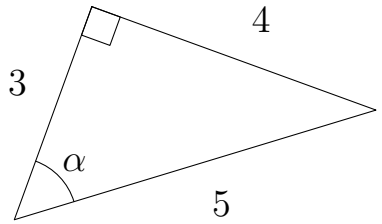
Example. Given the following triangle, find $\sin(\alpha)$, $\cos(\alpha)$, and $\tan(\alpha)$:



Example. Given the following triangle, find $\sin(\alpha)$:



Example. Given the following triangle, evaluate:

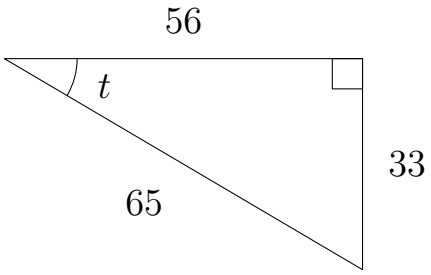


$\sin(\alpha)$ $\csc(\alpha)$

$\cos(\alpha)$ $\sec(\alpha)$

$\tan(\alpha)$ $\cot(\alpha)$

Example. Given the following triangle, evaluate:

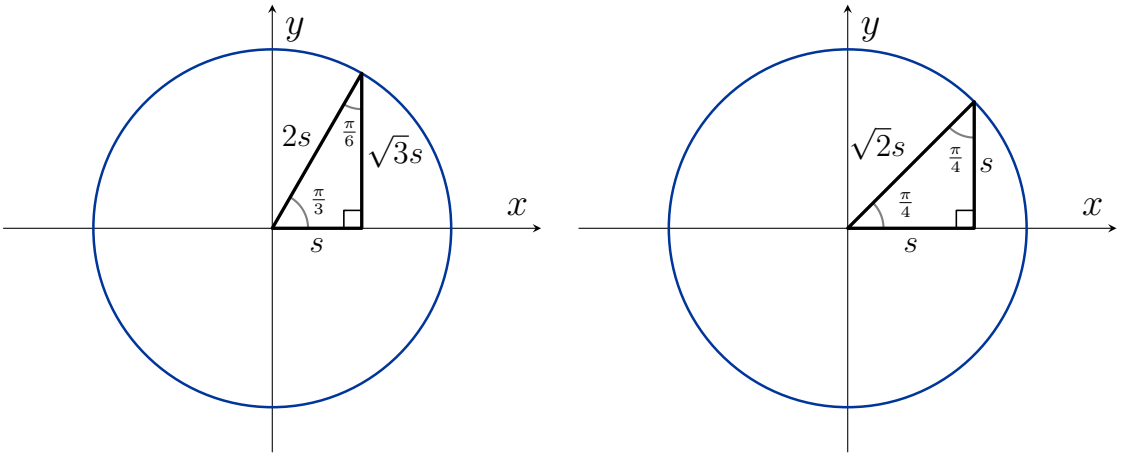


$\sin(t)$ $\csc(t)$

$\cos(t)$ $\sec(t)$

$\tan(t)$ $\cot(t)$

Side lengths of special angles:



Example. Find the exact value of the trigonometric functions of $\theta = \dfrac{\pi}{3}$, using side lengths.

$\sin(\theta)$ $\csc(\theta)$

$\cos(\theta)$ $\sec(\theta)$

$\tan(\theta)$ $\cot(\theta)$

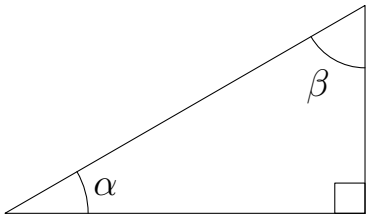
Example. Find the exact value of the trigonometric functions of $\vartheta = \dfrac{\pi}{4}$, using side lengths.

$\sin(\vartheta)$ $\csc(\vartheta)$

$\cos(\vartheta)$ $\sec(\vartheta)$

$\tan(\vartheta)$ $\cot(\vartheta)$

Equal cofunctions of complements:



$$\sin(\alpha) = \cos(\beta)$$
$$\sin(\beta) = \cos(\alpha)$$

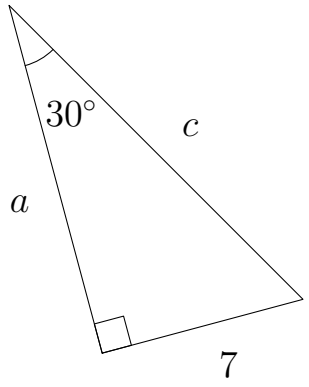
Definition. (Cofunction Identities)

$$\cos(\theta) = \sin\left(\frac{\pi}{2} - \theta\right) \qquad \sin(\theta) = \cos\left(\frac{\pi}{2} - \theta\right)$$
$$\sec(\theta) = \csc\left(\frac{\pi}{2} - \theta\right) \qquad \csc(\theta) = \sec\left(\frac{\pi}{2} - \theta\right)$$
$$\cot(\theta) = \tan\left(\frac{\pi}{2} - \theta\right) \qquad \tan(\theta) = \cot\left(\frac{\pi}{2} - \theta\right)$$

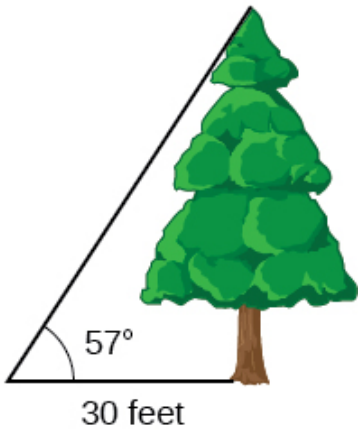
Example. If $\sin(t) = \frac{5}{12}$, find $\cos\left(\frac{\pi}{2} - t\right)$.

Example. If $\csc\left(\frac{\pi}{6}\right) = 2$, find $\sec\left(\frac{\pi}{3}\right)$.

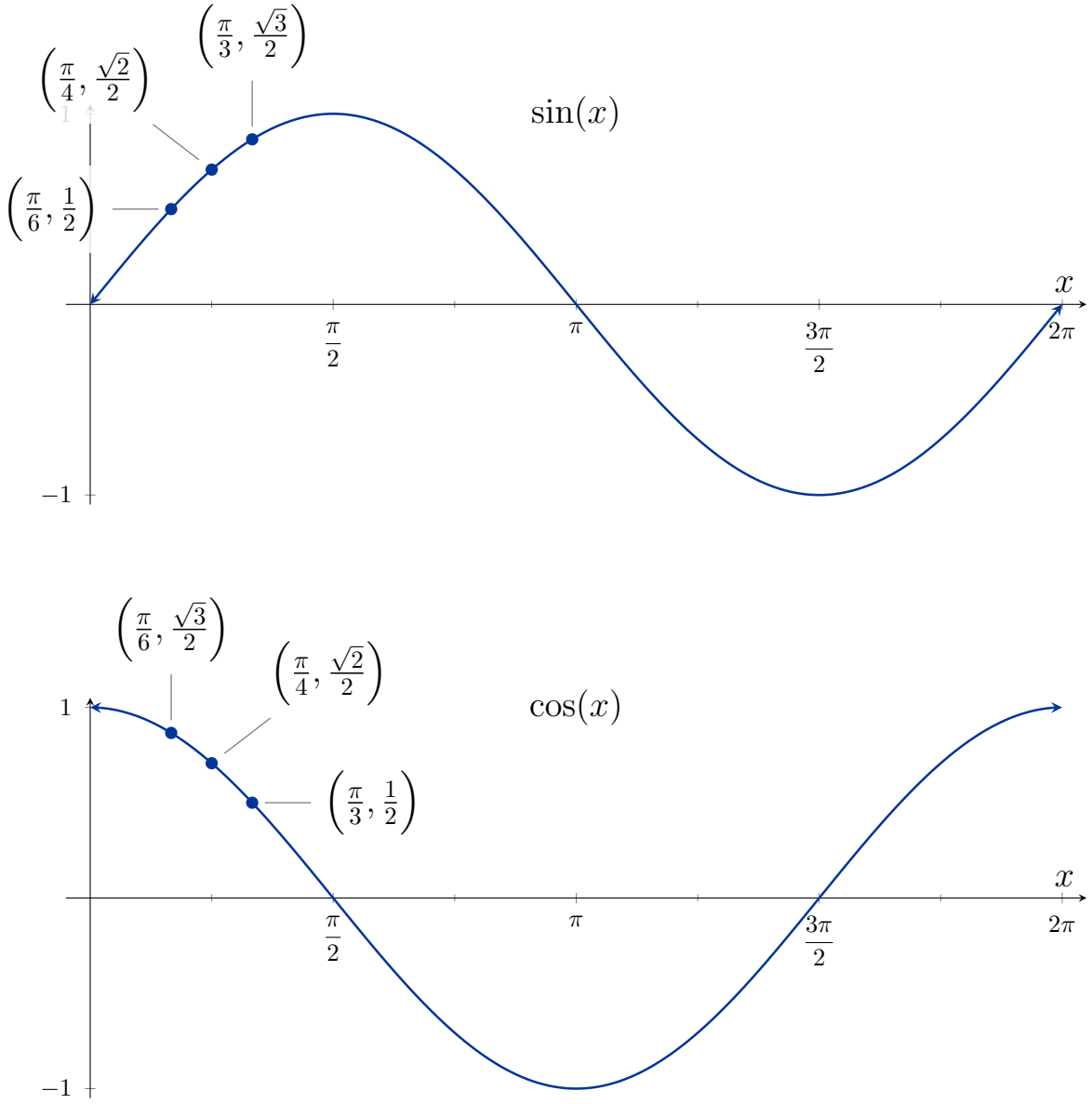
Example. Find the unknown sides of the following triangle:



Example. To find the height of a tree, a person walks to a point 30 feet from the base of the tree. She measures an angle of 57° between a line of sight to the top of the tree and the ground. Find the height of the tree.



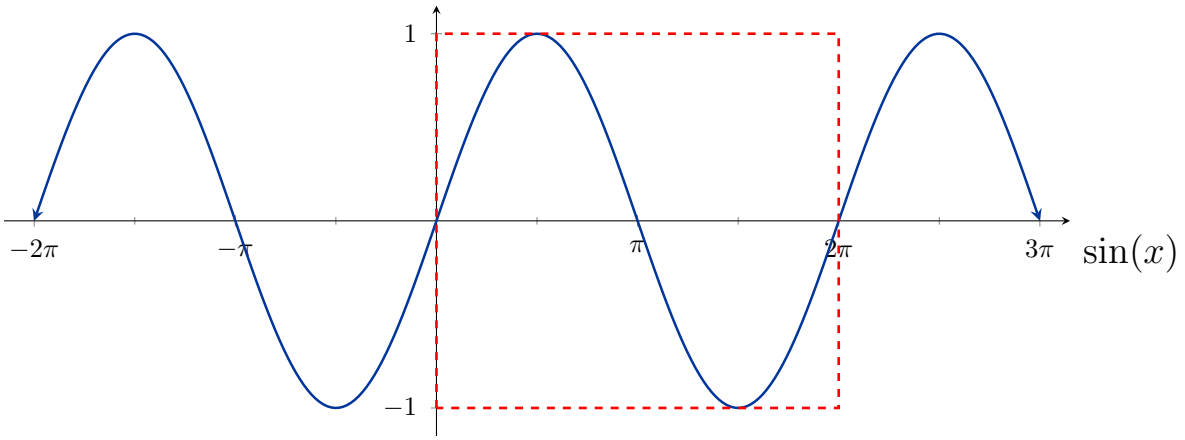
6.1: Graphs of the Sine and Cosine Functions

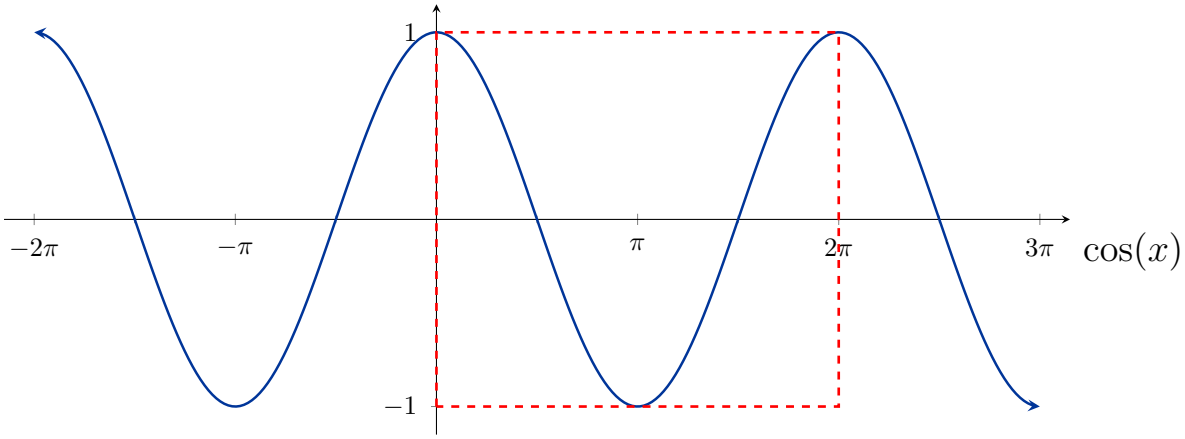


Animated graphs

Recall that

- $\sin(\theta)$ and $\cos(\theta)$ have period 2π,
- $\sin(\theta)$ is an odd function, and
- $\cos(\theta)$ is an even function:





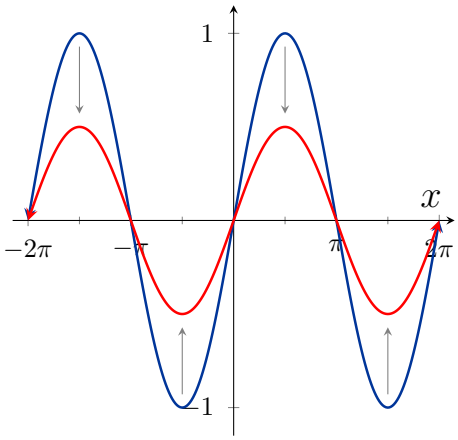
Recall from transformations of functions:

$$af(bx-h)+k$$

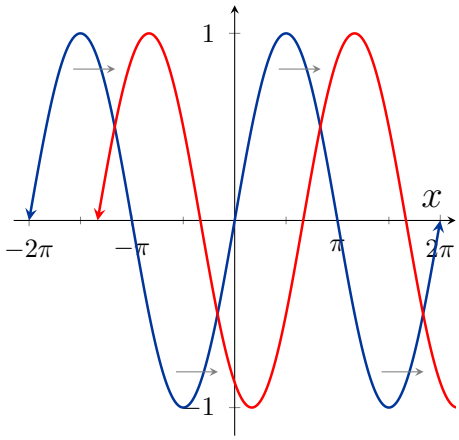
We have that $f(x)$ is

- stretched vertically by a
- stretched horizontally by $1/b$
- shifted right by h
- shifted vertically by k

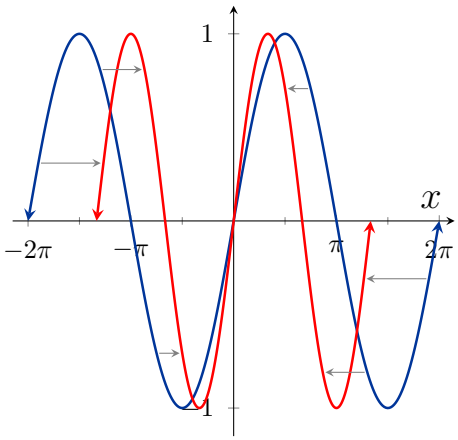
$\sin(x)$ vs. $\frac{1}{2}\sin(x)$



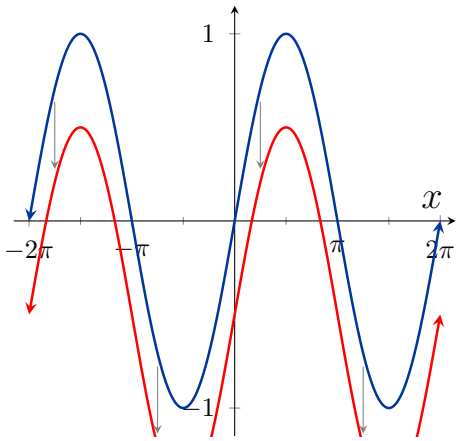
$\sin(x)$ vs. $\sin\left(x-\frac{2\pi}{3}\right)$



$\sin(x)$ vs. $\sin\left(\frac{3x}{2}\right)$



$\sin(x)$ vs. $\sin(x)-\frac{1}{2}$



Definition. (Sinusoidal functions)
A **sinusoidal function** is a function that has the same general shape as a sine or cosine function. The general forms of sinusoidal functions are

$$y = A\sin(Bx - C) + D \quad \text{and} \quad y = A\cos(Bx - C) + D,$$

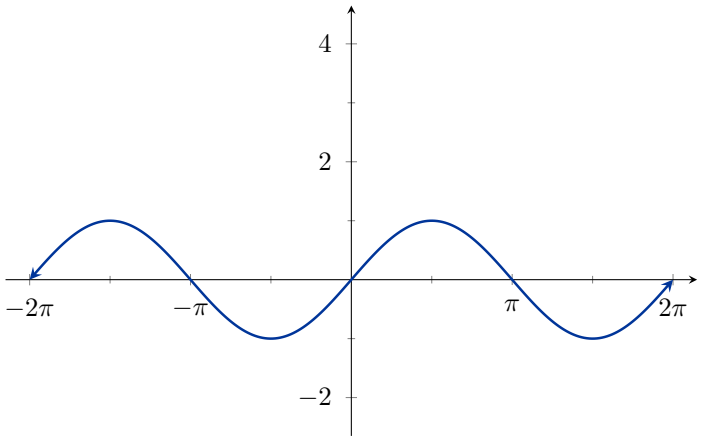
where

- A determines the amplitude,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

Example. Identify the amplitude, period, phase shift, and vertical shift of

$$3\sin(2x) + 1$$

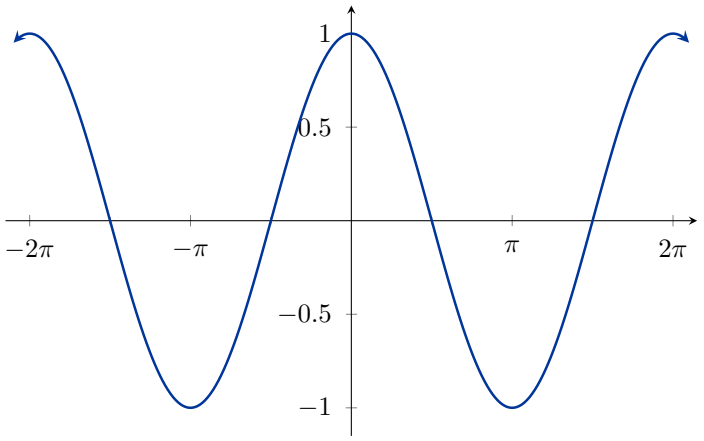
and sketch the graph:



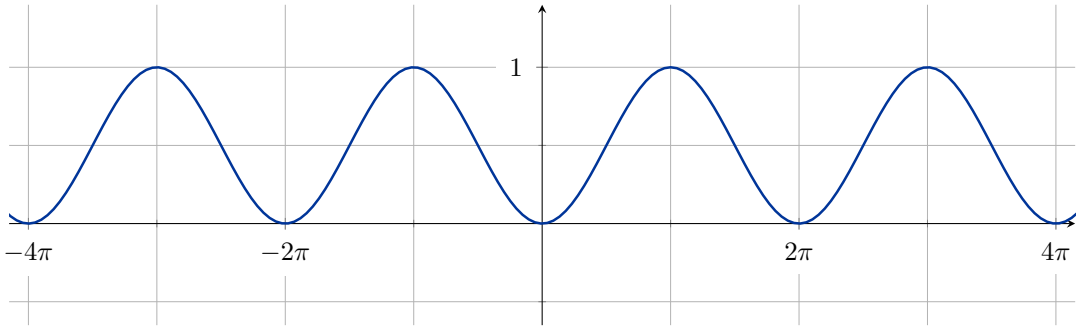
Example. Identify the amplitude, period, phase shift, and vertical shift of

$$\frac{1}{2}\cos\left(\frac{x}{3}-\frac{\pi}{3}\right)$$

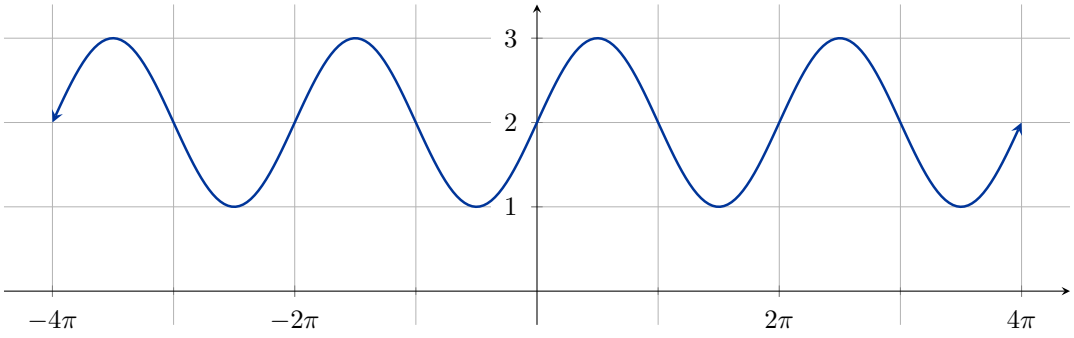
and sketch the graph:



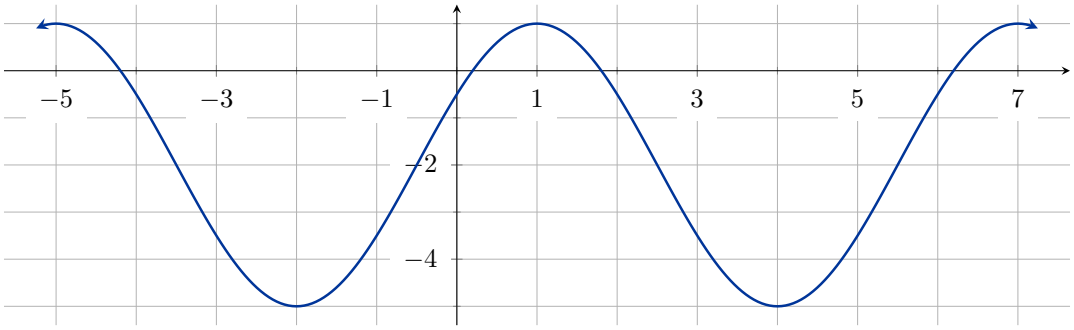
Example. Determine an equation for the following sinusoidal function:



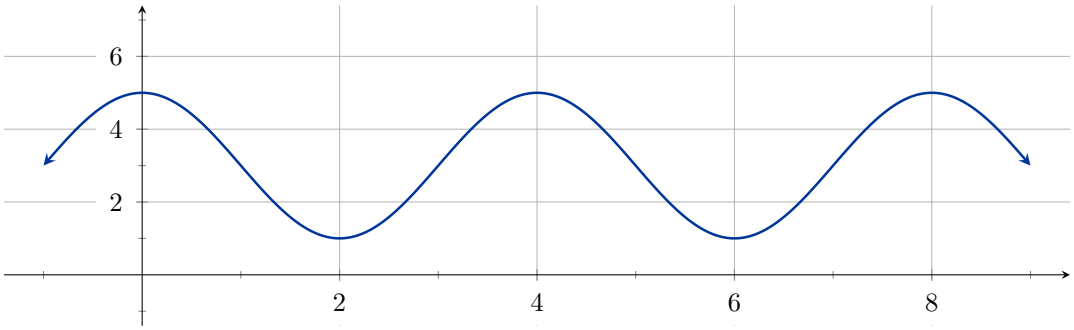
Example. Determine an equation for the following sinusoidal function:



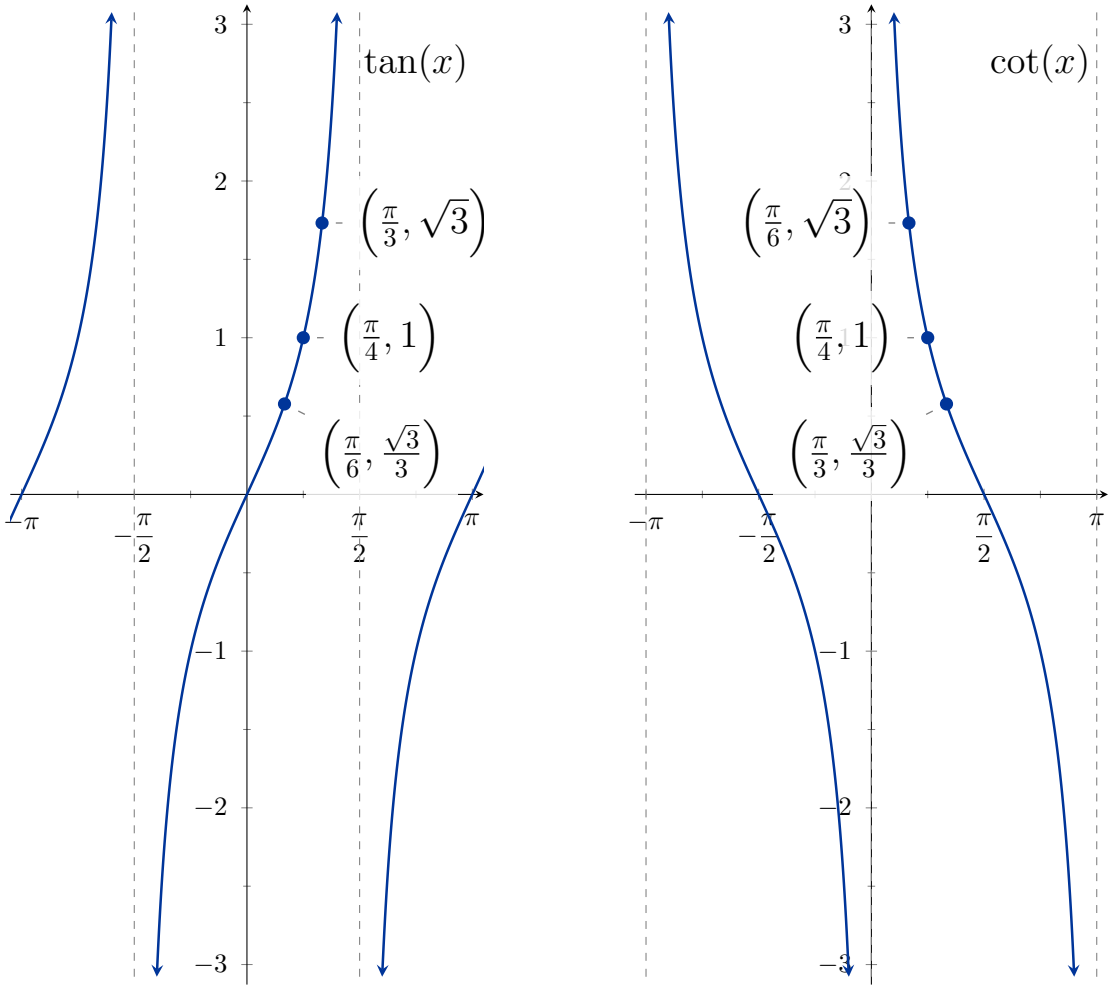
Example. Determine an equation for the following sinusoidal function:

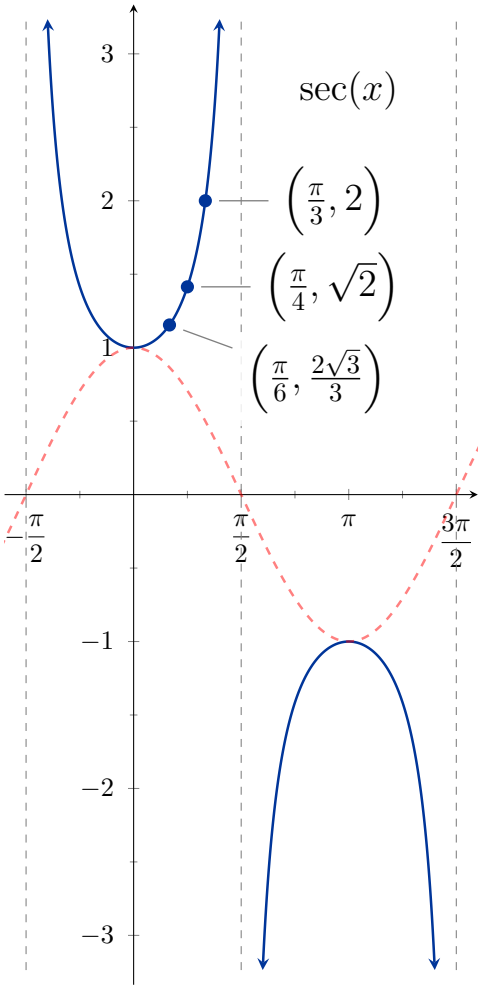


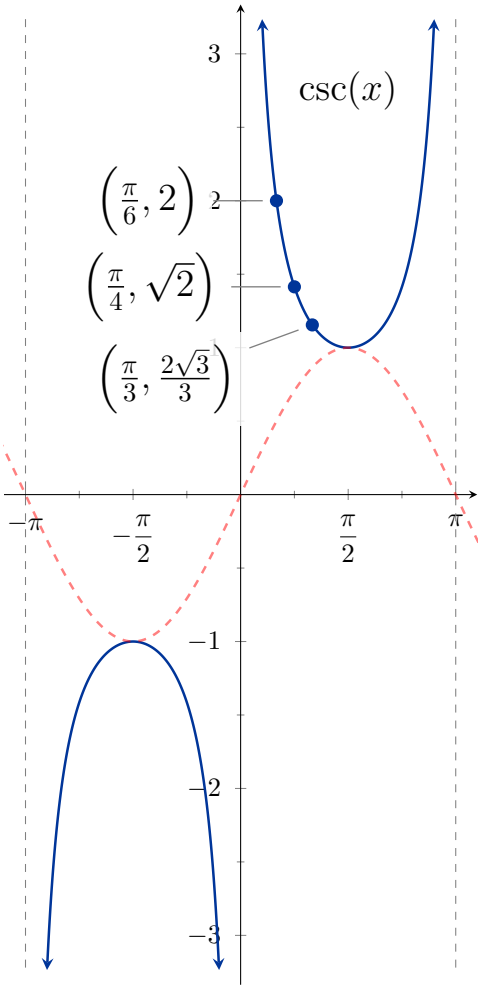
Example. Determine an equation for the following sinusoidal function:



6.2: Graphs of the Other Trigonometric Functions




sec(x)
3
2
1
−1
−2
−3
−π/2
π/2
π
3π/2
(π/3, 2)
(π/4, √2)
(π/6, 2√3/3)


csc(x)
3
2
−1
−2
−3
−π
−π/2
π/2
π
(π/6, 2)
(π/4, √2)
(π/3, 2√3/3)

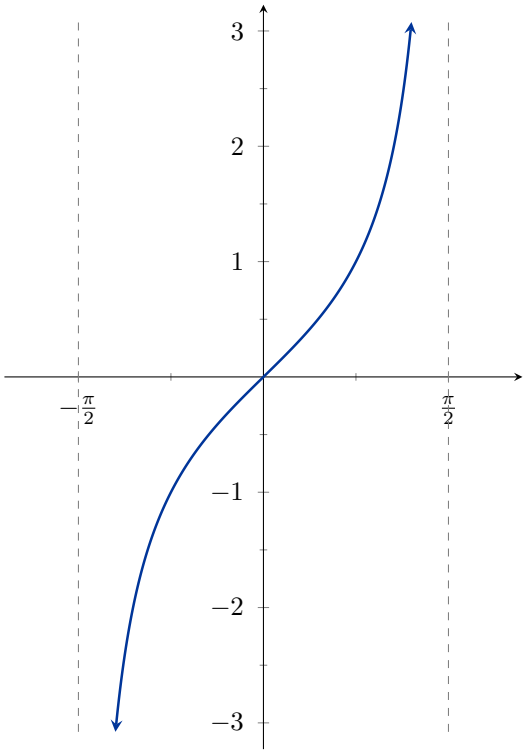
The general forms of the other trigonometric functions are

$$y = A\tan(Bx - C) + D, \qquad y = A\cot(Bx - C) + D,$$
$$y = A\sec(Bx - C) + D, \qquad y = A\csc(Bx - C) + D.$$

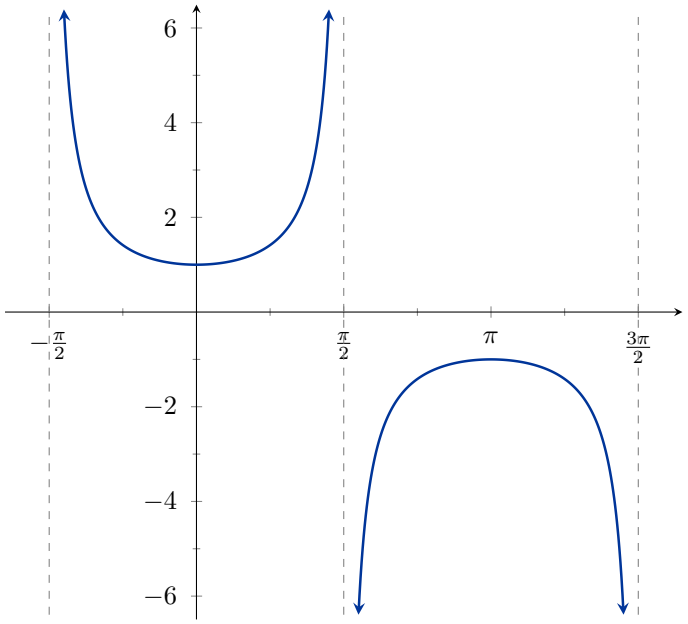
where

- A determines the scaling factor,
- B determines the period,
- C determines the phase shift, and
- D determines the vertical shift.

Example. Draw one period of the function $y = \frac{1}{2}\tan\left(\frac{\pi}{2}x\right)$



Example. Identify the domain, range, and draw one period of $y = 2.5 \sec\left(\frac{2}{5}x\right)$.



Example. Identify the domain, range, and draw one period of $y = -\cot\left(x + \frac{\pi}{2}\right)$.

